

SAPIENZA
UNIVERSITÀ DI ROMA

Maritime Trade and Conflict: An Empirical Study of Nigerian Regions

Facoltà di Economia
Master Degree in Economics and policy evaluation

Alessandra Amicucci
ID number 1940795

Supervisor
Prof. Valerio Leone Sciabolazza

Co-Advisor
Prof. Michele Di Maio

Academic Year 2024/2025

Thesis defended on 21/10/2025
in front of a Board of Examiners composed by:

Michele Di Maio (chairman)
Massimiliano Tancioni
Giuseppe Ragusa
Marco Ventura
Valeria Patella
Massimo Aprea
Milos Ciganovic

Maritime Trade and Conflict: An Empirical Study of Nigerian Regions
Sapienza University of Rome

© 2025 Alessandra Amicucci. All rights reserved

This thesis has been typeset by LaTeX and the Sapthesis class.

Author's email: email amicucci.1940795@studenti.uniroma1.it

Abstract

This thesis studies how exposure to international maritime trade shapes subnational conflict in Nigeria. For doing so, we built an exposure to trade index. We interacted changes in container port throughput (CPT) with the Liner Shipping Connectivity Index (LSCI) and the inverse distance to the nearest commercial port. Then, we studies the effect of the trade index to conflict, in particular ADM2 level events from ACLED (2009–2019). Using two-way fixed effects we found that higher exposure to trade is associated with a stabilizing effect on conflicts at the micro level, especially in poorer areas, while effects are near zero or weaker in more developed districts. Accounting for heterogeneity, results reported a stronger effect in the eastern coastal regions and for ports oriented toward vegetable goods and imports, and they attenuate in regions with abundance of water availability, a proxy for agricultural rents, aligning with the result of the baseline model. We continued the analysis estimating a falsification test using non-commercial ports and that supported our main hypothesis, demonstrating an effect of trade in conflict, rather than mere geographic proximity. The findings highlights the central role of pre-existing economic and spatial inequalities in mediating the stabilizing effects of globalization

Contents

1 Introduction **1**

2 Data **5**
2.1 Depend variable 5
2.2 Trade exposure 8
2.2.1 *Shock* 8
2.2.2 *Linear Shipping Connectivity* 9
2.2.3 *Distance to trade ports* 10
2.3 Control variable 12
2.4 Heterogeneity 13
2.4.1 *Geographical and sector characteristics* 13
2.4.2 *Tribes* 15
2.4.3 *Water availability* 16

3 Methodoloy **17**
3.1 Conceptual framework 17
3.2 Identification Strategy 18
3.3 Empirical Strategy 18
3.4 Endogeneity concerns 20

4 Results **22**
4.1 Baseline model 22
4.2 Heterogeneity 23
4.2.1 *Geography* 25
4.2.2 *Sector* 25
4.2.3 *Tribes* 26
4.2.4 *Water avaibility* 26

5 Conclusion **28**
5.1 Main findings 28
5.2 Interpretation 29
5.3 Limitations 29
5.4 Policy implications 29
5.5 External validity and future research 30
5.6 Contribution 30

A Conflict **31**

B Nightlights **33**

C Heterogeneity **35**

D Replication Code **38**

List of Figures

2.1 Nigeria conflicts 2001-2006-2009-2019 7
2.2 Conflict trend 7
2.3 LSCI and CPT trend 10
2.4 Comparative maps: location of ports(a) and minimum distance from them(b) 11
2.5 Distribuition of nightlights during 2009-2012-2015-2019 13
2.6 Spatial distribution of nightlight and high-NTL areas (2009–2019) . 14
2.7 Average coastal nightlights 15

A.1 Dominant conflict type 31

B.1 Conflict and nightlight 33
B.2 Confroning conflict, nightlight and different type of rent 34

C.1 est and ovest 36
C.2 log density of nightlights 36
C.3 Import and export 37
C.4 Vegetable and chemicals 37

List of Tables

2.1 Panel coverage and sparsity of conflict events 6
2.2 Mean conflicts by terciles of trade exposure 11

4.1 Correlation between port shocks and conflict: commercial vs all ports 22
4.2 Regional and sectoral heterogeneity (combined) 24
4.3 Water Availability: Levels & Heterogeneity 27

A.1 Panel coverage and sparsity of conflict events 32

C.1 Summary statistics by North/South macro-region 35

Chapter 1

Introduction

The study how international economic integration affects stability in a country remains one of the central question in development economics. A vast literature looks at the effects of trade liberalization on growth and welfare in advanced economies, yet evidence for developing countries remains limited (Dix-Carneiro & Kovak, 2023; Goldberg & Pavcnik, 2007) . In these settings, deep structural inequalities, weaker institutions, and a high reliance on primary commodities may shape very different transmission channels. In these contexts, trade represents an important channel through which external shocks influences local incomes and political stability. (UNCTAD, 2012). The existing empirical literature has mostly focused on commodity price shocks as the principal external driver of conflict dynamics, like the studies of Dube and Vargas (2013) and Brückner and Ciccone (2010). They show that international price changes can influence the likelihood of civil conflict, although the direction of the effect depends on whether a country is an exporter or importer of the resource, as well as on the quality of local institutions and the labor intensity of the commodity. Parallel studies investigate how wars and instability disrupt trade and integration, thus the reverse relationship (Rauschendorfer & Shepherd, 2022). Little evidence exists on how trade shocks themselves, rather than resource or conflict shocks, affect the probability of conflicts.
This analysiss attempts to be a first step in studying how exposure to trade influences subnational conflicts in Nigeria. Unlike previous studies that analyse price fluctuations or cross country variation in trade openness, the present analysis is based on data on port activity, maritime connectivity, and local socio-economic characteristics to measure how the diffusion of trade shocks across space affects the intensity of local conflict events. The aim is to move the focus from aggregate to micro variation, where the mechanisms can be observed more directly.

The effect of trade on conflict is theoretically mixed. As stated by Martin et al. (2008), that develop a theoretical framework in which trade openness can have different role in conflict depending on the relative strength of two opposing mechanisms: deterrence, which increases the cost of war through interdependence, and insurance, which reduces mutual dependence and can make conflict more tolerable. Empirical evidence supports both sides of this ambiguity. Berman and Couttenier (2015), on which this study is based,, shows that positive external income shocks can reduce

conflict because it could improve economic opportunities and raise the opportunity cost of violence. Conversely, trade may increase inequality or shift incentives, as discussed by Amodio et al. (2024). This means that the net effect of trade depends on local economic structure, the distribution of gains from trade, and the capacity of institutions to manage distributional tensions. Other mechanisms can influence this relationship that can influence the final effect of trade, such the type of goods traded and the local institutions, as shown by Dube and Vargas (2013) and Berman et al. (2017). In unequal economies, external shocks can therefore operate in two opposite directions: they may stabilize societies by smoothing consumption or destabilize it when benefits are accumulated by elites. We need spatial data to capure how trade exposure varies, and so, to identify which of these forces dominates. In this context, our analysis contributes by focusing on a distinct empirical channel: variations in maritime trade connectivity across Nigerian coastal regions. By examining localized exposure to global trade, the study explores how integration through ports may either stabilize or destabilize conflict dynamics, depending on local economic conditions, distributional patterns, and institutional capacity.

Nigeria has different characteristic that made it a compelling candidate for examining these dynamics, as it is a coastal, rich in natural resources, and regionally fragmented country whose economy depends on external trade. Its geography combines well connected port areas with vast inland regions only loosely integrated into national and global markets. These heterogeneities make the foundation that we need for assessing how trade liberalization could influences conflict through income shocks. Moreover, Nigeria exhibits other characterists useful for the study such as marked disparities between East and West regions, ethnic fragmentation, and a recurrent pattern of localized violence, which together amplify the potential for uneven economic transmission. Those channell will be disussed on chapter 3 and 4. In particular, Nigerian coastal regions host both mayor commercial ports and resource rich areas, wich is a naturl environment to develop exposure to trade and dependence on extraxting rents.
The Delta Niger region, for instance, has been characterized for long by violence linked by oil extranction and local grivience (Obi, 2010). At the same time Nigeria's conflict intensity and spatial heterogeneity, as reported in te study of Bertoni et al. (2019) make it a suistable context to study both the rapacity and opportunity cost channels through which trade may affect conflict dynamics.
The country, in general, exhibits substantial religious polarization and ethnic heterogeneity and strong disparities in income, education, and healthcare, producing a highly unequal social landscape (Montalvo & Reynal-Querol, 2005). In this setting, patterns of resistance have evolved over time—from nonviolent mobilization to militarized confrontation. In particolar Environmental degradation in the Niger Delta caused by petroleum pollution, together with federal control over resource revenues, has further fueled protests and episodes of violence, as reported by Campbell (2010). In general, coastal regions benefit from access to maritime infrastructure, while interior districts remain more isolated and vulnerable to economic exclusion. Understanding whether trade openness mitigates or exacerbates such disparities is crucial for interpreting the political economy of globalization in Africa.

To capture these mechanisms empirically, we developed a spatial trade exposure index. It combines three components that can link global trade dynamics to local geography: (i) trade shocks at the port level, measured as annual changes in port level Container Port Throughput (CPT) from UNCTAD; (ii) connectivity of the port to other international one, proxied by the Liner Shipping Connectivity Index (LSCI) from UNCTAD, which is used to quantify each port's integration into global liner networks; (iii) geographical proximity, expressed as the inverse of the minimum distance between each second level administrative unit (ADM2) and its nearest port.

In this specification, smaller distances and higher connectivity imply greater exposure to international trade shocks. The resulting index reflects the idea that trade shocks propagate unequally across space, because it depends on the local access to maritime infrastructure and the integration into global shipping networks (Berman & Couttenier, 2015; Dube & Vargas, 2013). This structure ensures that only areas which are geographically and economically close to ports experience significant variation in exposure when trade conditions change. We constructed the index analysing the interaction between time varying national portual trade shocks (represented by $CPT \times LSCI$) with fixed spatial heterogeneity (that is the inverse distance), constructed this was the index respect the validation request by Goldsmith-Pinkham et al. (2020). The idea beyond is to captures how fluctuations in global maritime activity translate into differential local effects across Nigeria's territory in line with the exposure measures used in recent empirical work on trade and conflict (Berman & Couttenier, 2015).
A higher value of the index indicates that a district is more exposed to international trade shocks transmitted through Nigeria's maritime network. This exposure can be interpreted economically as a measure of both opportunity and risk: regions that are closer to more connected ports are more affected by changes in global shipping activity. In peripheral areas, this integration into trade networks could raise the opportunity cost of violence, fostering stability (Berman & Couttenier, 2015). At the same time there could be the opposite mechanism in areas that present more resources. In these regions, trade may amplify competition over rents or reinforce existing inequalities, potentially generating new sources of tension (Dube & Vargas, 2013; Michalopoulos & Papaioannou, 2013). The spatial variation represented by the proximity to ports, is a key measurement for identifying heterogeneous effects across Nigeria's territory, because coastal and port proximate districts are likely to react differently from remote inland areas. The index aim to provide a lens through which we can examine how globalization's benefits and risks are transmitted unequally within a country allowing us to investigate whether the access to international trade serves as a stabilizing force by increasing local opportunities, or as a destabilizing factor when economic gains are concentrated and unequally shared.

Building on this framework, we fist linked the constructed exposure index to subnational conflict data from ACLED, focusing on consumption related events that best reflect the opportunity cost mechanism of conflict dure the 2009–2019 period. Then, we estimated a series of fixed effects models aiming to isolate the effect of trade exposure from time invariant and spatial characteristics and finally, we explore heterogeneity across regions, sectors, and environmental conditions, examining how

local structures mediate the impact of trade integration on stability.

Overall, our findings show that greater exposure to international trade is associated with a reduction in conflict, particularly in less developed and more isolated districts, but the stabilizing effect weakens or is reversed in those regions when we consider the distribuition of environmental resources, suggesting that the benefits of trade depend largely on how its gains are distributed. These findings contribute to the debate on the political economy of globalization, showing that trade can act as a force for stability when embedded in inclusive local development.

The contribution of this study is double. First, it aim to provide a subnational evaluation of trade shocks and conflict in Nigeria, integrating spatially detailed datasets on trade flows, port infrastructure, connectivity, conflict events, and environmental resources. This perspective allows the analysis to move beyond national averages and uncover within country heterogeneity in the economic and social consequences of trade. Second, the results reveal that the impact of trade liberalization is highly heterogeneous. Exposure to maritime trade tends to reduce conflict incidence less developed districts, but can reinforce instability in wealthier or rent dependent areas, where new rents or inequalities emerge. This pattern highlights the central role of pre-existing economic and spatial inequalities in mediating the stabilizing effects of globalization. Beyond its empirical contribution, the thesis also seeks to build a conceptual bridge between two strands of literature that have largely evolved in parallel: studies of external income shocks and conflict (Amodio et al., 2024; Berman & Couttenier, 2015) and analyses of spatial exposure to global markets inspired by gravity models (Anderson & van Wincoop, 2003). By linking these frameworks through a data driven, geography based measure of trade integration, the study provides new evidence on how international linkages permeate domestic political economies.

The remain of the thesis is organized as follows.Chapter 2 describes the data sources and the construction of key variables. Chapter 3 presents the conceptual framework and empirical strategy. Chapter 4 reports the main empirical results and chapter 5 concludes by summarizing the main findings.

Chapter 2

Data

In our empirical analysis, we combined data from different sources to derive a panel of 179 coastal subnational geographical (ADMIN2) units for the period from 2009 to 2019.

2.1 Depend variable

Conflict. Data on conflict event are drawn from the "Armed Conflict and Location Event" ACLED (2025) which cover conflict event from 1999 to 2024. The dataset provides the exact location of events, the disorder type, and other characteristics related to the conflict. (Raleigh et al., 2010)
In constructing the dataset, we focused on consumption related, non violent, conflict events rather than rapacity based ones. This choice was guided by both theoretical and empirical considerations given that standard trade models usually predict that trade openness influences a country's income in a positive way. However, globalisation has not always reduced inequalities in emerging countries, causing unequal distribution (Dix-Carneiro & Kovak, 2023). As stated by Dube and Vargas (2013), an increase in income can reduce conflict, but on the other hand, higher income means there is more to fight over these resources. The existence of two opposing influences, opportunity cost and rapacity, suggests that some shocks may mitigate conflict, meanwhile other shocks could exacerbate it, depending on the relative strength of the two effects. The study of Amodio et al. (2024) shows that exposure to exports increasess simultaneously economic activity and political violence and McGuirk and Burke (2020) have shown that, if there is a positive price shocks, producers' revenues increase but consumers' real incomes decrease, enhancing the probability of protests or demonstrations. Conversely, Berman and Couttenier (2015) find that at the micro level positive income external shock, such us those generated by the trade, tend to have a pacifyng effect on conflict. In general, we can differenciate consumption related mechanisms as more directly influenced by exogenous trade shocks, while violent conflicts over resources as affected by different dynamics, often linked to the strategies of armed groups. From an empirical point of view, limiting to consumption related events also mitigates concerns about endogeneity: resource predation can in itself influence local trade flows, risking to reflect both cause and

effect simultaneously.
The indipendent variable therefore measures the annual number of consumption related conflict events for ADM2, that have been filtered to retain only: (i) subcategories of events not related to predation, such as protests, demonstrations, arrests, agreements or territorial changes not directly linked to resource predation. (ii) Events reported in narrative notes as disputes over consumer goods or livelihoods (keywords such as land, livestock, pasture, farm, grazing, oil, gold, looting). Events clearly associated with predatory motives (e.g., looting, armed clashes over resource extraction, kidnappings for ransom) are excluded.

After filtering, we limited the dataset to events occurring in coastal regions, then projected them into a consistent spatial reference system and overlaid them on Nigeria's second level administrative units (ADM2). For each ADM2-year, the count of non predatory conflict events is available, which serves as the outcome variable in the regression analysis.
Table 2.1 reports the summary statistics for the conflict variable. It shows that

Table 2.1 Panel coverage and sparsity of conflict events

	Value
Number of ADM2 units	179
Years covered	2009–2019
Observations (ADM2 year)	1,980
Share of zero conflict observations	0.698
Median number of conflicts	0
90th percentile	3
99th percentile	17
Mean number of conflicts	1.01
Standard deviation	3.04

Note: The outcome variable counts annual consumption related conflict events from ACLED at the ADM2 level.

70% of ADM2–year cells record zero events, indicating that violence is spatially and temporally concentrated. This fragmentation motivates the focus on within district variation and the inclusion of fixed effects.

Figure 2.1 displays the spatial distribution of consumption related conflict events across Nigerian coast in the years 2009, 2012, 2015 and 2019. The maps show a marked increase in conflicts between 2009 and 2019. In 2009, events were concentrated in a few clusters, while in 2019 they spread across most coastal regions, gradually extending westwards and inland. In 2015, there was a much more significant increase compared to 2012, and was greater than the gap between 2012 and 2009, suggesting an exponential growth dynamic.

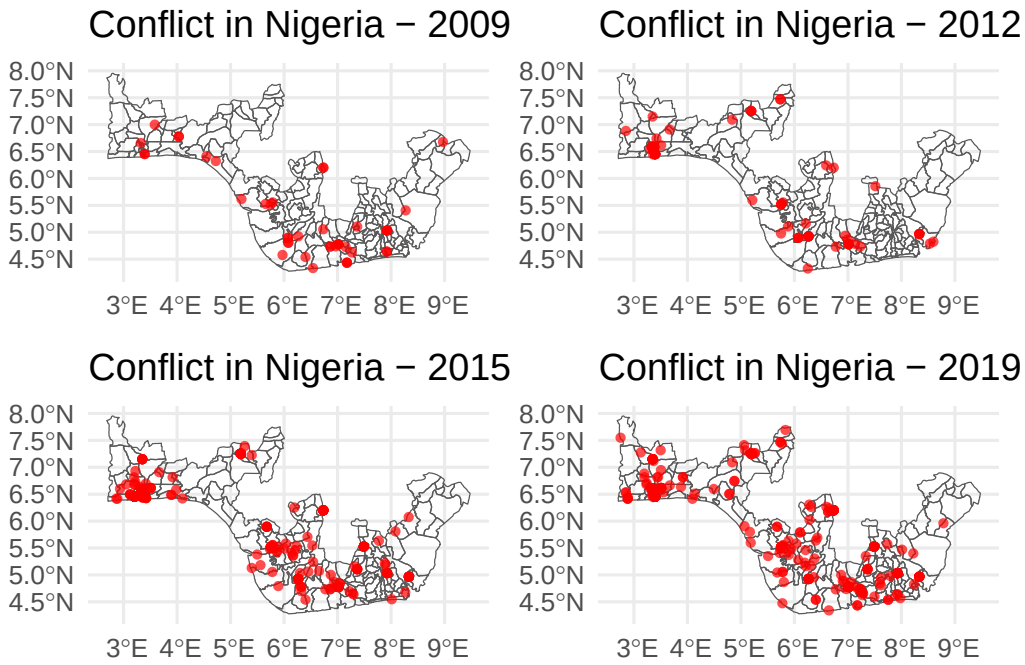


Figure 2.1 Nigeria conflicts 2001-2006-2009-2019

Figure 2.2 shows the trend in the total number of conflict events per year (sum at ADM2 level). Focusing on the period 2009-2019, we can see that in the early years the number of incidents remained relatively low, but there has been a sharp increase since 2012, with progressive growth leading to a peak between 2015 and 2019. This trend confirms what has already been highlighted by the maps, namely a pattern of intensification and spread of conflicts over the decade.

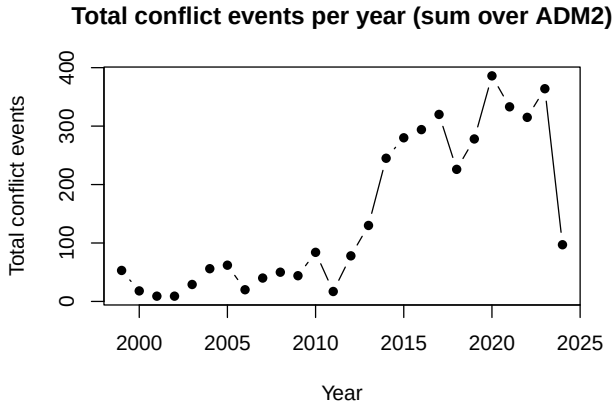


Figure 2.2 Conflict trend

2.2 Trade exposure

As a second step we proceed building an index of trade exposure. This measure combines the temporal variation of a national trade shock with the spatial heterogeneity of local exposure, ensuring that both time and geography contribute to identifying variation.
The index was constructed based on the intuition that a trade exposure shock can significantly influence the dynamics of conflicts. In particular, we consider how the impact of trade shocks varies geographically depending on the distance from trade ports: as highlighted by Berman and Couttenier (2015), regions closer to ports are in fact more exposed to external shocks. Following this intuition, we constructed an index that combines the measure of national connectivity with a local proxy for geographical accessibility to ports.

$$Index = Shock \times LSCI \times 1/(distminnorm + \varepsilon) \tag{2.1}$$

Here, the shock is represented by Container Port Throughput (CPT), a dataset compiled by UNCTAD that estimates the total number of containers handled, by country, expressed in twenty-foot equivalent units (TEU); LSCI stands for Liner Shipping Connectivity, which measures a country's integration into global liner shipping networks, also from UNCTAD; finally, there is the subnational variable, which is the inverse of the normalized minimum distance to the nearest commercial port and ε is a very small positive constant introduced to avoid division by zero.

The main idea is that being close to commercial ports makes it easier to get to international markets. In other words, areas near ports are more exposed to international trade, while areas far away are relatively isolated. This insight is consistent with empirical evidence that regions closest to ports are most affected by global shocks, whether positive or negative, that are transmitted through trade (Berman & Couttenier, 2015; McGuirk & Burke, 2020). Similar to the quasi experimental approach by Amodio et al. (2021), who exploited Israel's 2008 ban on dual use imports to the West Bank as an exogenous shock to trade, the present index captures how sudden or structural changes in trade integration may affect local economic and conflict dynamics. While their evidence shows that trade restrictions depress wages and increase violence, here the focus is to assess whether greater maritime accessibility and trade intensity can mitigate conflict in a developing country context such as Nigeria.

2.2.1 *Shock*

Foreign trade shock is measured using Container Port Throughput (CPT), a dataset compiled by UNCTAD (2025b) that reports the annual number of containers handled in each country's ports, expressed in TEU. CPT is used as an indicator of the intensity of maritime trade flows, as most international trade, in terms of volume, takes place by sea (Ayesu & Boateng, 2024). Meanwhile the LSCI captures the potential capacity of a national maritime transport network, the CPT reflects actual containerised traffic and thus provides a more direct measure of exposure to global

supply and demand shocks. Empirical studies confirm that changes in CPT have tangible macroeconomic effects. For example, Ayesu and Boateng (2024) demonstrate that increased container traffic can boosts employment in African economies with maritime ports, highlighting its role as an engine of economic development. At the same time, the OECD/ITF report by Ducruet et al. (2014) points out that throughput also reveals structural vulnerabilities: in ports where efficiency is low, high levels of throughput can amplify the impact of external shocks due to congestion and longer periods of stay. From a policy perspective, UNCTAD (2012) recognises CPT as a key indicator of port development performance in emerging economies, highlighting its value as a standard measure of exposure to international trade. In this analysis, we consider the CPT as an exogenous source of variation in maritime trade flows which, interacting with measures of national connectivity (LSCI) and local port accessibility (distance from port), captures the heterogeneous transmission of global trade shocks to subnational regions.

2.2.2 *Linear Shipping Connectivity*

The national component of the index is represented by the Liner Shipping Connectivity Index (LSCI), developed by UNCTAD (2025a) and frequently used in maritime trade literature as shown in Mishra et al. (2021) and Fugazza and Hoffmann (2017). The LSCI measures a country's structural capacity to integrate into global trade, based on variables such as the number of liner services, active operators, ships employed, ship size and total capacity in Twenty Foot Equivalent Units (TEU). It is internationally recognised as an indicator of trade accessibility and competitiveness, particularly relevant in developing countries where maritime transport is the main route for international trade. Empirical studies confirm that greater maritime connectivity reduces trade costs and promotes growth: for example, Fugazza and Hoffmann (2017) estimate that the absence of direct connections can reduce the value of bilateral exports, while the Review of Maritime Transport UNCTAD (2012) shows that improvements in the LSCI ranking are systematically associated with higher trade volumes and greater market access.
Figure 2.3 presents the trend of the LSCI and CPT in Nigeria over the period 2009–2019. The two indicators show a similar trend: sustained growth until 2014–2015, followed by a contraction phase and a partial recovery in recent years. Their co-movement confirms that LSCI and CPT capture complementary dimensions of the same trade dynamic: the former reflects the structural capacity for integration into global trade, the latter the actual measure of the volume of containers handled. For this reason, both are included in the construction of the trade shock index used in the empirical analysis.

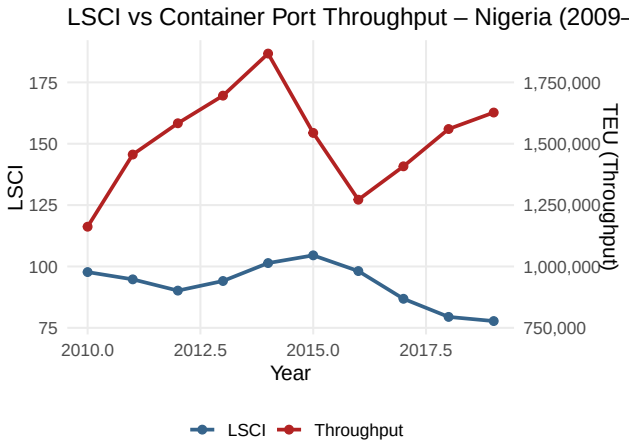


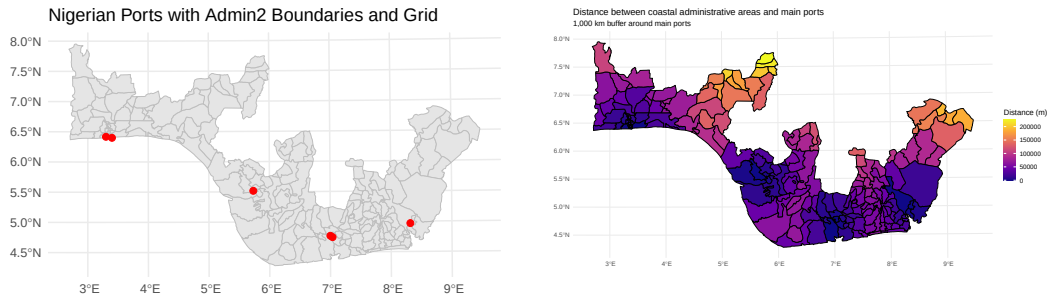
Figure 2.3 LSCI and CPT trend

2.2.3 *Distance to trade ports*

The role of distance from ports is crucial to understanding the heterogeneity of trade shocks. Following the intuition of gravity models Anderson and van Wincoop (2003), geographical distance remains a central determinant of trade costs and access to markets. In our framework, minimum distance from ports is therefore employed as a measure of local exposure to international trade shocks (Anderson & van Wincoop, 2003). According to Berman and Couttenier (2015), regions located closer to major seaports are more directly integrated into international trade networks and therefore more strongly affected by external income shocks. Their findings show that positive shocks reduce the likelihood of conflict within these open regions, while the effect largely disappears in remote inland areas. This pattern suggests that better access to trade increases local incomes and raises the cost of using violence, thereby mitigating conflict. Following this idea, we incorporated distance from ports into the construction of the trade index. The logic beyond this choice is that Nigerian districts closer to seaports are more exposed to changes in global trade flows and should therefore exhibit a stronger response to consumption related conflicts in the event of external shocks than districts located further inland.
Nigeria have ten seaports overall; however, both the Nigerian Ports Authority and IMF PortWatch identify six as the country's principal commercial gateways (International Monetary Fund & University of Maryland, 2025; Nigerian Ports Authority, 2025). Accordingly, our main distance measure is computed to the nearest of these six ports: Lagos, Tin Can Island, Port Harcourt, Onne, Warri, and Calabar.

Figure 2.4a shows the position of the six main ports overlapped for the ADM2 subnational regions. These ports correspond to those officially listed by the Nigerian Ports Authority and identified in the IMF PortWatch database as handling the largest shares of national maritime trade activity (International Monetary Fund & University of Maryland, 2025). For each coastal ADM2, the distance from the nearest port was calculated, converted into kilometres and then normalised through

a min–max transformation. To ensure comparability across units and to give greater weight to areas close to ports, the normalised distance was inverted so that higher values indicate stronger exposure to trade.
Figure 2.4b report the spatial distribution within a 1,000 km buffer. We can see that the coastal areas immediately adjacent to the ports, in dark purple, are naturally much more accessible, while the distance increases rapidly as one moves inland, highlighted by the lighter shades. This geographical heterogeneity forms the basis for the variable, which assigns greater exposure to areas closer to ports and progressively reduces that of more remote regions.



(a) Main trade ports **(b)** Buffer around main ports

Figure 2.4 Comparative maps: location of ports(a) and minimum distance from them(b)

We grouped ADM2 units by terciles of trade exposure and it reported a u shaped association, as shown in Table 2.2, suggesting that conflict incidence is lowest at intermediate levels of exposure and higher at both extremes. This pattern suggests the coexistence of stabilizing opportunity income effects and destabilizing rent seeking dynamics, further explored in the empirical analysis.

Table 2.2 Mean conflicts by terciles of trade exposure

Exposure tercile	Mean conflicts	Mean index value
1 - Low exposure	1.41	−360,370
2 - Medium exposure	0.68	30.5
3 - High exposure	1.38	605,912

Note: ADM2 units are grouped by terciles of the trade exposure index. Conflict incidence is lowest in the middle tercile, suggesting a non linear relationship between exposure and instability.

While the LSCI index measures Nigeria's overall capacity to engage in global maritime trade, the minimum distance to ports is a reflection of regional disparities in the actual access to these connections. By integrating these two dimensions with the external shock represented by CPT, the index reflects both the macro structure of the port system and local differences in exposure. In this way, not only are

general changes in the country's position within maritime networks captured, but also how such shocks are absorbed differently depending on geographical proximity to ports. The combination of these elements allows us to move from a uniform national perspective to a more detailed one, highlighting the territorial heterogeneity of the effects of maritime trade and their relevance to the dynamics of conflicts.

2.3 Control variable

Nighlights Night time lighting (NTL) data are widely used in the literature as an indicator of economic activity, especially in contexts where conventional economic statistics are scarce or unreliable. The underlying idea is that artificial nightlight is strongly correlated with income levels, infrastructure and access to energy, thus serving as a reliable indicator of local development (Amodio et al., 2024; Chen & Nordhaus, 2011; Henderson et al., 2012). In the case of Nigeria, nightlights have been used to assess access to electricity and regional inequalities, confirming their strong association with patterns of socio-economic development. To construct this variable, we relied on the Harmonised Global Night time Light Dataset (1992-2018) developed by Li et al. (2020), which combines the DMSP-OLS series with the latest VIIRS observations into a consistent time series. The harmonisation procedure corrects for satellite drift, blooming, and sensor inconsistencies, thus ensuring temporal comparability. The dataset is available in yearly composites and can be aggregated spatially at multiple scales, making it particularly suitable for integration with ADM2 level units.For each coastal ADM2 and for each year, we calculated the logarithm of nightlight intensity to reduce asymmetry and facilitate interpretation in regressions.

Figure 2.5 shows the spatial distribution of nightlight intensity in Nigeria's coastal ADM2s. Light concentration is clearly clustered around Lagos, Port Harcourt, and other industrial centres, while rural areas show very limited brightness, consistent with the literature on energy poverty in Nigeria.

In addition, we defined a dummy variable equal to one for ADMs with values above the 75th percentile of the logaritim distribution of the nightlights within the year. Choosing a relatively high threshold ensures a stronger contrast between these highly developed localities and the remaining, less illuminated areas that are more rural. This distinction allows us to test whether trade shocks have differential effects between more and less economically developed regions. As illustrated in Figure 2.6, light concentration is clustered around the main port cities, while large portions of the coastal hinterland remain darker.

Moving on to temporal dynamics, Figure 2.7 tracks the average intensity of nightlights over the period 2000–2018. The trend reveals an overall increase in luminosity, reflecting the gradual urbanisation of Nigeria, but also a significant decline around 2013 and 2016. These fluctuations are consistent with the technical issues documented in the harmonised dataset. In particular, Li et al. (2020) notes that the decline observed in the period 2015–2016 is also present in independent VIIRS composites and attributes it to sensor fluctuations and the transition from

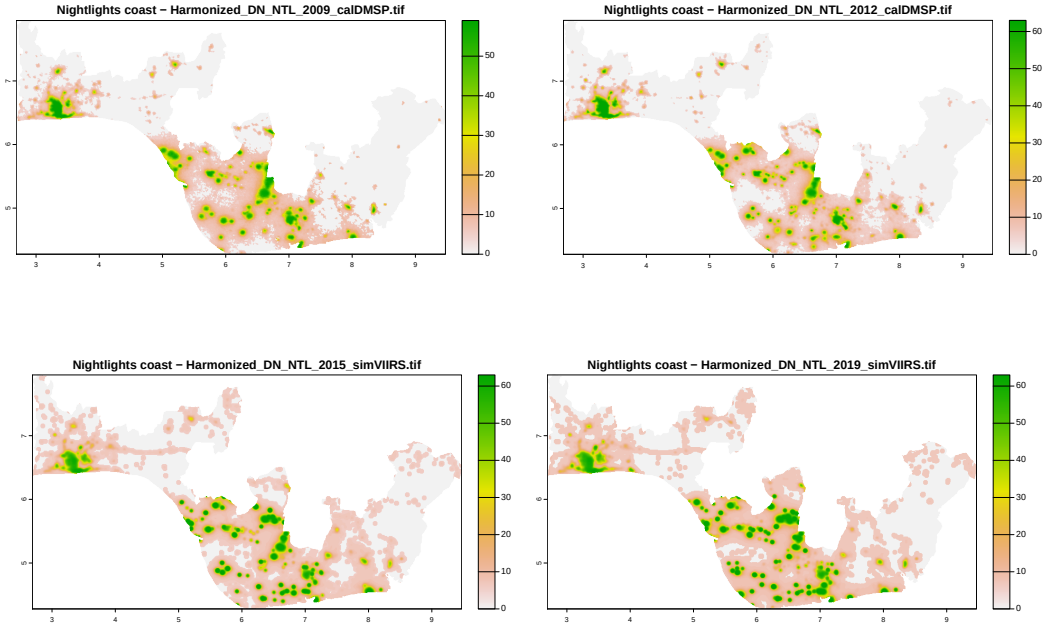


Figure 2.5 Distribuition of nightlights during 2009-2012-2015-2019

DMSP to VIIRS rather than an actual economic contraction. Similarly, Islam (2024) highlights the challenges posed by overglow and sensor calibration in the Nigerian context, emphasising that apparent discontinuities in time series may reflect data artifacts rather than real changes in development. It is important to note that such noise has been highlighted in previous studies as a limitation of remote sensing for socio-economic monitoring, but the harmonisation procedure mitigates much of the distortion. We analyse the fluctuations in Appendix B
In summary, nightlights provide a proxy for local economic activity. The expectation is that areas with lower nightlight intensity (i.e. poorer and less developed regions) should be more sensitive to foreign trade shocks, as consumption disruptions are more likely where households face tighter constraints on basic goods and services.

2.4 Heterogeneity

2.4.1 *Geographical and sector characteristics*

To account for spatial and structural differences across Nigeria's coastal economy, we classified ADM2 units and ports along two complementary dimensions. The heterogeneity framework follows recent work emphasizing that the effects of economic integration depend on pre existing local characteristics such as geography, sectoral specialization, and social composition (Berman & Couttenier, 2015; McGuirk & Burke, 2020).
First, East–West regions were defined according to the median longitude of ADM2

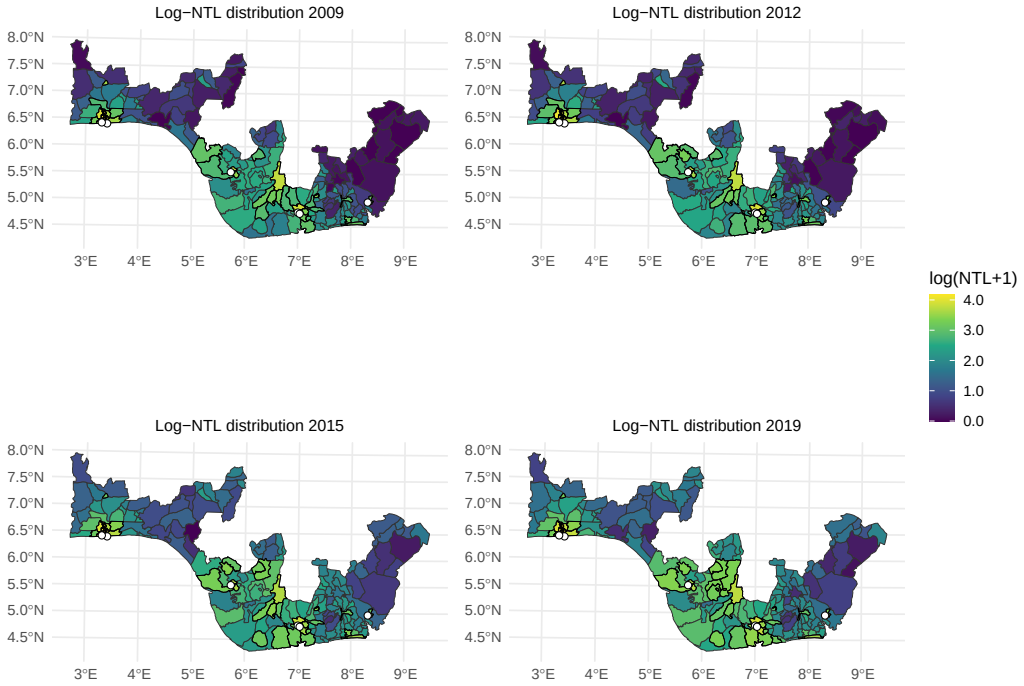


Figure 2.6 Spatial distribution of nightlight and high-NTL areas (2009–2019)

Note: The colour scale represents the logarithm of nightlights intensity (log(NTL+1)). Administrative boundaries are drawn in grey; thick black borders indicate ADM2 units above the 75th percentile of luminosity (High-NTL areas). White dots denote the main Nigerian ports (Lagos, Tin Can Island, Port Harcourt, Onne, Warri, and Calabar).

centroids, distinguishing the eastern coastal belt that is centered around Port Harcourt and Calabar, from the western belt dominated by Lagos and its surroundings. This partition captures historical and economic disparities in port activity and industrial development. Similar geographical differences have been studies across other African economies, where proximity to ports and market access are key determinants of local development and conflict dynamics (Michalopoulos & Papaioannou, 2018; Storeygard, 2016). Following Henderson et al. (2012) and Amodio et al. (2024), nightlights are used as a consistent proxy for economic activity to capture these spatial differences.

Second, sectoral and trade flow specializations were derived from the IMF PortWatch database, which provides information on the predominant categories of goods traded through each port (International Monetary Fund & University of Maryland, 2025). Following the data availability since 2019, ports were classified as either vegetable oriented or chemical-oriented, the two main product groups after minerals, and as predominantly import or export oriented depending on the share of incoming and outgoing containerized flows. These distinctions are used later in the

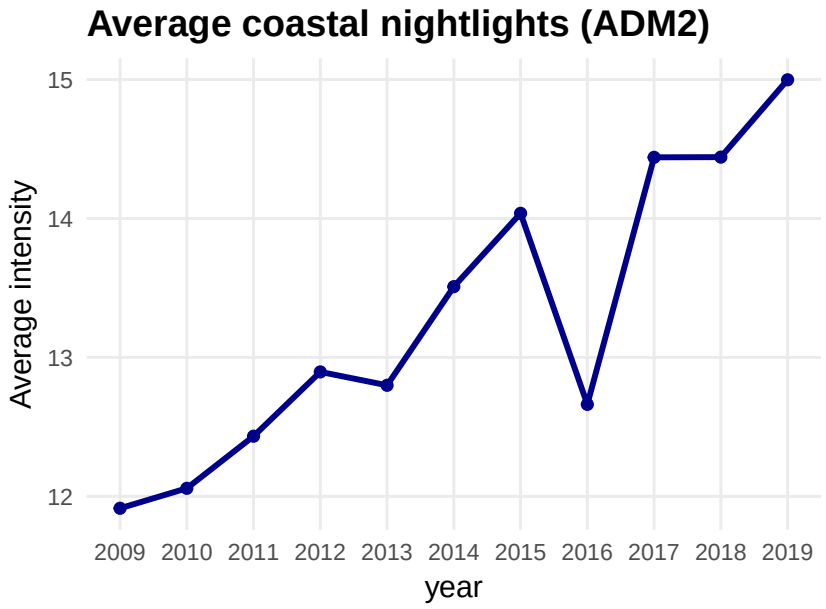


Figure 2.7 Average coastal nightlights

heterogeneity analysis to test whether the pacifying effect of trade exposure differs by regional position, sectoral composition, or trade direction.

2.4.2 *Tribes*

Tribal composition is used as a proxy for ethnic diversity across Nigerian regions. Previous studies, such as Fearon (2003) and Montalvo and Reynal-Querol (2005), show that high ethnic polarization tends to increase the probability of conflict bcause it weakens social trust and collective action. Contrary, ethnic fragmentation may reduce the risk of conflict by preventing the dominance of a single group. This approach follows a long tradition of studies linking ethnic structure to political instability and local conflict (Fearon & Laitin, 2003; Michalopoulos & Papaioannou, 2013).
Building on this idea, we construct two variables capturing tribal heterogeneity: (i) the share of the dominant tribe in each ADM2 (the maximum share among local tribes), and (ii) the fractionalization index, defined as:

$$T_i = 1 - \sum_i s_i^2 \tag{2.2}$$

where s_i is the population share of each tribe. This index ranges from 0 (one tribe dominates) to 1 (many equally sized tribes), thus reflecting the degree of local diversity. We then analyse how these social structures interact with economic conditions to shape the effect of trade shocks on conflict incidence.

2.4.3 *Water availability*

We expect that regions with greater water availability are those where economic activity is higher and more connected to national markets. Since the vegetable sector represents one of the main traded categories in Nigerian ports, second or third depending on the port, water resources are an essential input and thus a relevant proxy for agricultural income. For this reason, we use water availability as an indicator of agricultural activity and local economic potential (FAO & IIASA, 2020). The inclusion of this variable follows evidence that climatic and resource endowments influence both agricultural productivity and conflict dynamics in Africa (Burke et al., 2015). The data come from FAO–GAEZ, which provides raster information on the mean quantity of water resources (or hydrological balance) available in each geographical unit. We calculate the mean value for every ADM2 area, obtaining a spatially continuous and relatively stable characteristic of the territory. To facilitate heterogeneity analyses, districts are ranked by this indicator and then split in three tercile dummies: low, below 33%, medium, between 33% and 66%, and high, above 66%. This variable is then used to test how the pacifying effect of trade varies with the underlying resource environment.

Summary descriptives illustrating these geographical, sectoral, and social patterns are presented in Appendix C, providing preliminary evidence of marked regional disparities and heterogeneity in local economic conditions.

Chapter 3

Methodoloy

We expect the economic impact of trade shocks to vary across different ADM2 units, with areas most exposed to international trade experiencing fewer consumption related conflict events. Following this intuition, the index presented in the chapter Data was constructed, combining two national components that vary over time with a fixed spatial component. This approach is consistent with consolidated practices in the literature that combine aggregate shocks with subnational heterogeneity (Autor et al., 2013).

The mechanism linking trade exposure and conflict is consistent with the opportunity cost and grievance framework discussed by Amodio et al. (2021). Their evidence suggests that economic integration and access to trade can shape local conflict dynamics by altering individuals' economic incentives and perceptions of welfare. In this thesis, the focus is on the opposite mechanism: instead of trade restrictions, we study whether higher trade openness and proximity to maritime ports, which expand local economic opportunities, can mitigate the incidence of violence. The same channels highlighted by Amodio et al. (2021) are therefore expected to operate in reverse, with improved trade connectivity reducing both grievance and the opportunity to engage in conflict.

3.1 Conceptual framework

To better capture the mechanisms through which trade exposure affects social stability, we construct a specific subset of conflict events reflecting consumption or income related tensions rather than resource seeking or political violence. Starting from the ACLED dataset (ACLED, 2025), we select events classified as protests or demostrations, the types less likely to have a link with resource predation. In particular, we identify income related events through a keyword based text filter, selecting those that mention terms associated with basic needs or livelihood shocks, such as food, fuel, price, subsidy, market, ration, maize, rice, bread, or tax. Conflict incidence can be represented as (3.1) where Y_{it} captures opportunity income effects (the effect of trade in income, reducing conflict) and R_{it} captures rent seeking incentives (the opposite effect of trade that incentivises conflict).

$$Conflict_{it} = f(Y_{it}, R_{it}) \tag{3.1}$$

Our index captures exogenous variation in the income effect transmitted through trade shocks, holding the incentives largely constant by excluding rapacity related events.

3.2 Identification Strategy

Identification relies on within district (ADM2) variation in exposure to national trade shocks over time. Because the trade shock (CPT $\times$ LSCI) is common across units while the distance is time invariant, only the interaction between these components generates differential effects once ADM2 and year fixed effects are included. In our case, the CPT was delayed by one year to mitigate possible simultaneity issues, while the LSCI represents the degree of integration of the Nigerian port system into global maritime networks. Spatial heterogeneity is introduced through the minimum distance from ports, which serves as a proxy for accessibility to international markets. Following the intuition of (Berman & Couttenier, 2015), regions located closer to ports are more directly integrated into international trade networks and therefore more strongly affected by external shocks. In our context, this motivates the use of minimum distance from ports as a measure of local exposure, assuming that trade shocks are transmitted more intensely to areas with better access to maritime infrastructure. In this sense, the empirical design is consistent with the validity conditions discussed by (Goldsmith-Pinkham et al., 2020)
The empirical model also includes both spatial (ADM2) and temporal (year) fixed effects. In this way, given the same national shock, only differences in proximity to ports generate variation between units, allowing the heterogeneous effect of trade shocks on conflict to be isolated. To account for spatial heterogeneity the distances from ports where normalised. Since the relationship between trade exposure and conflict is expected to decrease with distance, the raw distances (measured in kilometers) were first inverted to capture proximity rather than remoteness. To ensure comparability across regions, the proximity measure was then rescaled between 0 and 1, so that values closer to 1 represent areas with higher accessibility to maritime trade. This transformation preserves relative spatial differences while avoiding scale distortions that could bias the exposure index.

3.3 Empirical Strategy

All models are estimated using OLS with two-way fixed effects (ADM2 and year). Standard errors are clustered at the ADM2 level to account for serial correlation within units over time.
The empirical strategy consists of three steps.

First, we estimate the basic model in two specifications: an aggregate version, in which the exposure index is considered in its overall form Equation(3.2), and a second forma that take into account the heterogeneity, which distinguishes administrative units based on the intensity of economic development measured by nightlights Equation (3.3). This allows us to analyse both the average effect of the trade shock and the differences between more and less developed areas.

$$Conflict_{it} = \beta Index_{it} + \gamma log(Nightlights_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \tag{3.2}$$

$$Conflict_{it} = \beta_1 Index_{it}^{HigNL} + \beta_2 Index_{it}^{LowNL} + \gamma log(Nightlights_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \tag{3.3}$$

The equations 3.2 and 3.3 estimate the effect of trade exposure on local conflict incidence. In the baseline form, equation 3.2 we regressed conflict events in administrative units i and years t on the trade exposure index. We also controlled for local economic activities, proxied by nightlights as described in the Chapter Data, and both unit and fixed effect (respectively μ_i and λ_t). In this way, we are are to capture the variation overtinme whitin unit eliminating the effect of unobserved variables that remain constant overtime, as geographical characteristic, and common temporal shocks affecting all region in a given year.

We estimated equation 3.3 to study the effect taking into account the heterogeneity across ADM2 with different level of local development, proxied by the concentration of nightlights. By doing so, the model test whether the pacifying effect, or the destabilizing one, of trade is stronger in more or less developed.

Both models are estimated using Ordinary Least Squares (OLS) with two-way fixed effects and clustered standard errors at the ADM2 level, following standard practice in subnational conflict analyses (e.g., Berman and Couttenier (2015) and Dube and Vargas (2013)). This approach ensures consistent estimates by exploiting temporal variation within regions while accounting for spatial correlation and unobserved heterogeneity.

Second, we perform a falsification test to verify that the results do not simply reflect a spurious correlation between conflict and geographical proximity to port infrastructure. To this end, we reconstruct the index to exclude the six main trade ports in Nigeria, not just commercial ones. The idea is that if the effect were linked exclusively to physical proximity to a port, it should also emerge with this broader measure. Conversely, if the association is really driven by trade shocks, the results should only appear when considering ports that are actually relevant to international trade.

Third, beyond the baseline model, we explore whether the trade–conflict relationship varies across key structural dimensions. Following the idea that the effects of economic integration depend on pre-existing territorial and social characteristics (Berman & Couttenier, 2015), we analyse the heterogeneity of the effects across region, sector and environment.
In particular, we distinguish between (i) eastern and western regions, (ii) the predominant type of industry traded, (iii) the predominant trade flow, (iv) tribal dominance and fractorization and (v) water availability. These specifications allow us to investigate the differentiated transmission of trade shocks depending on the structural conditions of the territories. Equation 3.4 extends the baseline model to account for heterogeneity across structural or environmental dimensions (the first three categories).

$$Conflict_{it} = \beta_1 Index_{it} + \beta_2 (Index_{it} \times H_i) + \gamma \log(Nightlights_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \quad (3.4)$$

The interaction term captures whether the effect of trade exposure differs depending on regional location or sectoral specialization. In this model H_i is an indicator for the subgroup representing the heteroeneous variable, e.g. Export, Vegetables, etc.). The coefficient β_1 capture the effect of the trade exposure index when $H_i = 0$, while β_2 measure the effect when $H_i = 1$. A negative and statistically significant coefficient on β_2 would indicate that the pacifying effect of trade is stronger in the corresponding subgroup.

$$\begin{aligned} Conflict_{it} = {} & \beta_1 (Index_{it} \times LowNL_i) + \beta_2 (Index_{it} \times HighNL_i) \\ & + \beta_3 (Index_{it} \times LowNL_i \times T_i) + \gamma \log(Nightlights_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \end{aligned} \quad (3.5)$$

Equation 3.5 extend the baseline model taking into account heterogeneity from both tribal dominance and fractorization considering the economic differences across ADM2. The interaction term β_3 captures whether the effect of trade exposure on conflict in less developed regions depends on the degree of ethnic homogeneity or diversity. In particular, the variable T_i alternately represent the dominant tribal share, measuring the extent to which a single ethnic group constitutes the majority of the local population, and the ethnic fractionalization index, which captures the probability that two randomly chosen individuals belong to different groups. A positive and significant coefficient would indicate that the stabilizing effect of trade weakens as tribal dominance increases, suggesting that trade may exacerbate tensions where one group is predominant.
Conversely, a negative value would imply that trade is more stabilizing in more homogeneous territories.

$$\begin{aligned} Conflict_{it} = {} & \beta_1 Index_{it} + \beta_2 (Index_{it} \times HighWater_i) + \beta_3 (Index_{it} \times MediumWater_i) \\ & + \beta_4 (Index_{it} \times LowWater_i) + \gamma \log(Nightlights_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \end{aligned} \quad (3.6)$$

Equation 3.6 account for water availability heterogeneity using tercile indicators. In this analysis the coefficients β_1 is the trade–conflict slope in low water areas, $\beta 2$ and $\beta 3$ capture how the relationship between trade and conflict differs in medium and high water regions, suggesting that environmental can influence how economic shock are trasmitted locally.

3.4 Endogeneity concerns

A key methodological concern in estimating the basiline model and its variants is the potential endogeneity between trade exposure and conflict incidence.
In principle, trade shocks can influence local stability, but at the same time conflict itself may disrupt trade flows, transport infrastructure, leading to biased estimates if not properly addressed,m suggesting that causality may run in both directions.

Moreover, omitted variables correlated with both trade and conflict, such as local political stability, could further bias the estimates.

To mitigate these risks, the analysis uses exogenous variation in maritime trade exposure, constructed from global shipping indicators, that are largely determined outside the local context and includes ADM2 and year fixed effects.
More specifically, the trade index combines CPT and LSCI at the port level and is then weighted by the inverse distance to each ADM2 unit to aim to ensures that most of the observed variation arise from global shocks rather than local conflict dynamics.

Chapter 4

Results

In what follows, we present three sets of results. First we studied the baseline model estimating the effect of the trade exposure index in conflict taking into account the economic heterogeneity. Second we show the model taking into account the geographical, sectorial and tribal heterogeneity. Third, we investigate the effect of the index across region with different water availability.

4.1 Baseline model

Table4.1 reports the estimates of the main coefficient from Equation (3.2) and (3.3).

Table 4.1 Correlation between port shocks and conflict: commercial vs all ports

	(1) Commercial ports	(2) Commercial ports	(3) All ports	(4) All ports
Trade Exposure Index	-1.01e-07		-9.22e-09	
	(6.30e-08)		(2.1129e-08)	
*High Nightlight Intensity		-1.12e-07		-3.12e-08
(1 = Above the median)		(7.80e-08)		(2.85e-08)
*Low Nightlight Intensity		-5.61e-08***		7.30e-09
(1 = Below the median)		(5.43e-09)		(7.51e-09)
Nightlight intensity	-0.879***	-0.878***	-0.902***	-0.901***
	(0.249)	(0.248)	(0.259)	(0.260)
Observations	1,611	1,611	1,611	1,611
Time fixed effect	YES	YES	YES	YES
District fixed effect	YES	YES	YES	YES
R^2 (Adj.)	0.662	0.662	0.661	0.662

Fixed effects: ADM2 and year. SE clustered at ADM2 level.
High/Low Nightlight Intensity are ADM2 groups defined by the sample median of nightlights (harmonized DMSP-OLS/VIIRS). Entries in group rows report the simple slope of the Trade Exposure Index within the indicated group; group dummies themselves are absorbed by ADM2 FE. 358 observations dropped due to missing RHS variables. Signif.: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

It shows the results of the baseline specification, comparing the commercial ports with the broader set of Nigerian ports. Column (1) and (2) focus exclusively on the commercial ports, while Columns (3)–(4)exclude them as a falsification test.

Starting from Column (1), results indicates that the aggregate coefficient on the exposure index is small and not not statistically significant. However, when the sample is split to study the heterogeneity between high and low nightlight areas, as in Column (2), a strong heterogeneity emerges. This division was made to distinguish ADM2 units by local economic development, proxied by night time lights (as detailed in the Data section): where the areas with higher income are the ones with values above the 75 percentile (Amodio et al., 2024; Henderson et al., 2012).
In the areas with reatevely low levels of nightlights, the effect of the commercial shock is negative and statistically significant, suggesting that a greater exposure maritme trade reduces the likelihood of consumption based conflicts. Quantitatively, the coefficient implies that the trade index reduce conflict by 6 percent.
In contrast the coefficient is close to zero in the more developed area. This model is coherent with the intuition that the external trade shocks have stronger effect where the local economies are less diversified and more vulnerable to changes in global market integration (Berman & Couttenier, 2015). Thus distance from commercial ports is particularly important in the less developed regions, amplifying the regulatory effect of the trade shocks. This finding is especially relevant given that less developed areas are also more likely to have conflict (Brückner & Ciccone, 2010; Collier & Hoeffler, 2004)

Columns (3)–(4) serve as a falsification test. If the relationship was determined solely by proximity with any port, not only the commercial one, we would expect similar effects when Nigerian trade ports are excluded. Columns 3 and 4 extend the analysis, excluding the main trade ports. The results show weaker and not statistically significant coefficients, both for the aggregate and the splitted samples. The absence of an effect reinforces the interpretation that the mechanism is specificaly linked to international trade flows, rather than mere geographic proximity.

Overall, the results suggest that the link between trade and conflict is neither mechanic nor omnipresent, it rather depends commercial relevance of ports and the level of local economic development. The coefficients support the idea that international integration can provide stabilising benefit in the peripherical regions, and so the less developed ones economically speaking, while its marginal impact in more developed areas remains limited.

4.2 Heterogeneity

After establishing the baseline relationship, we explore whether the estimated effects vary across Nigeria's main macro regions and according to local social fragmentation. Table 4.2 combines these dimensions of heterogeneity: Panel A shows how the trade conflict relationship differs across regions, sectors and tribal fominance and fractorization, whereas Panel B reports the corresponding nightlight averages, illustrating the underlying economic disparities between groups.

Table 4.2 Regional and sectoral heterogeneity (combined)

Panel A. Index effects

	East vs West	Chemicals vs Vegetables	Export vs Import	Tribal	
East (slope)	-1.63e-07 *				
	(8.24e-08)				
West (slope)	-7.45e-09				
	(4.76e-08)				
Vegetables (slope)		-1.63e-07 *			
		(8.24e-08)			
Chemicals (slope)		-7.45 e-09			
		(4.76e-07)			
Export (slope)			2.13e-09		
			(3.29e-08)		
Import (slope)			-2.55e-07 ***		
			(4.87e-08)		
Low Nightlight Intensity				5.55e-05	-5.7e-08***
(1 = Below the median)				(4.41e-05)	(5.43e-9)
High Nightlight Intensity				-3.48e-07	-1.63e-08
(1 = Above the median)				(2.05e-07)	(-9.34e-09)
Dominance x Low NTL				5-53 e-05	
				(4.13e-05)	
Dominance x High NTL				5.16e-07	
				(2.77e-09)	
Fractionalization x Low NTL					3.88e-05
					(2.87e-05)
Fractionalization x High NTL					3.42e-07
					(2.45e-07)
Controls					
Nightights intensity	-0.884 ***	-0.884 ***	-0.884 ***	-0.877***	-0.877***
	(0.2493)	(0.2493)	(0.2470)	(0.248)	(0.248)
Observations 1,611	1,611	1,611	1,611	1,611	1,611
Time fixed effect	YES	YES	YES	YES	YES
District fixed effect	YES	YES	YES	YES	YES
Adj. R^2	0.665	0.665	0.670	0.662	0.662

Panel B. Average nightlights (mean / median / sd)

Group	Mean NL	Median NL	SD NL	N
East	9.78	6.57	10.4	979
West	16.8	8.91	18.9	990
Chemicals	16.5	8.41	19.2	957
Vegetables	10.3	6.76	10.6	1,012
Export	16.4	9.17	17.9	1,199
Import	8.48	5.83	9.52	770

Panel A Fixed effects: ADM2 and year. SE: clustered at ADM2 level. Rows labeled "(slope)" report the slope of the Trade Exposure Index within the stated group, estimated via specifications that allow group-specific slopes. High/Low Nightlight Intensity are ADM2 groups defined by the sample median of nightlights (harmonized DMSP-OLS/VIIRS). Tribal indices are time-invariant; their levels are absorbed by ADM2 FE, so only interaction driven slopes are identified. Signif.: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.
Panel B Nightlights (NTL) are mean annual radiance from harmonized DMSP-OLS/VIIRS. Statistics are computed at the ADM2 level over 2009–2019 (Mean/Median/SD across years within ADM2; N = number of ADM2 in each group).

4.2.1 *Geography*

Columns (1) investigates geographic heterogeneity: the index is interacted with East-/West dummies defined by the median longitude of coastal ADM2 centroids (details in Appendix C). The estimated effect is more negative and statistically stronger in the eastern belt—areas that are poorer and more connected to maritime activity, consistent with a vulnerability to trade mechanism. A detailed discussion of the spatial correlation between the macro regional partition, together with corresponding maps, is reported in Appendix C.

These patterns mirror the spatial distribution of economic development (Panel B). Nightlight data shows that western ADM2s exhibit higher and more dispersed nightlight values, largely driven by the Lagos metropolitan area, while eastern units are uniformly dimmer, reflecting a more rural and less industrialized economic structure. In this case, the gap between richer and poorer areas (as proxied by nightlights) is particularly pronounced. Detailed maps, descriptive statistics of nightlights by region, and additional robustness checks are reported in Appendix C

4.2.2 *Sector*

Columns (2)-(3) reports the sector heterogeneity: the index is interacted with two different sets of dummies: (i) the predominant type of industry traded through each port, either chemical or vegetable products (the two main sectors after minerals, based on data from International Monetary Fund and University of Maryland (2025), available since 2019 and used here as a proxy); and (ii) the predominant trade flow orientation, that is, whether the port handles mainly imports or exports (also based on data from IMF Portwatch). The estimated effect is negative and statistically significant for vegetables and imports. For presentation purposes, we refer to the coefficients as index_*veg* and index_*chem* (or index_*exp* and index_*imp*), where the former represents the baseline effect and the latter includes the interaction term. The stronger and statistically significant effects observed for vegetable and import oriented ports suggest that not only proximity and structural economic conditions matter, but also the nature and direction of trade flows. The results are consistent with previous findings that trade in basic consumption goods, such as agricultural products, can mitigate conflict by stabilizing local prices and incomes ((McGuirk & Burke, 2020)). The effect for vegetable oriented ports is also consistent with what is shown by Amodio et al. (2024): regions with lower nightlight intensity, typically linked to agricultural activity, are more vulnerable to economic shocks, suggesting that the welfare channel of trade is particularly relevant in such contexts. Similarly, import flows—particularly of essential goods—may enhance welfare and reduce consumption related tensions (Brückner & Ciccone, 2010; Dube & Vargas, 2013). Overall, these patterns reinforce the idea that trade integration contributes to stability primarily through welfare improving, rather than resource based, channels. Results are aligned with the underlying economic structure summarized in Panel B, where vegetable and import oriented ports correspond to areas exhibiting more dispersed nightlight distributions, a pattern typically associated with more rural and

less industrialized economies. Although the import/export distinction also reflects structural differences in local economies, import oriented ports show a systematically stronger effect than all other categories. Overall, these patterns reinforce the idea that trade integration contributes to stability primarily through welfare improving, rather than resource based, channels.

4.2.3 *Tribes*

Columns (5)–(6) report the results for the heterogeneity by ethnic composition.
Information on tribal distribution is used to proxy local ethnic diversity. Two indicators were constructed: tribal dominance and tribal fractionalization.
Tribal dominance identifies areas where a single ethnic group constitutes more than 50 percent of the local population (Esteban et al., 2012), while tribal fractionalization measures the probability that two randomly selected individuals belong to different ethnic groups (Collier & Hoeffler, 1998, 2004).
Empirical studies suggest that these two dimensions have opposite effects on conflict: Esteban et al. (2012) find that higher fractionalization tends to reduce conflict, whereas Collier and Hoeffler (2004) show that dominance may fuel conflict through capture of local rents and exclusion of minorities.
In our estimations, both coefficients are positive but not statistically significant, indicating no robust relationship between trade exposure and conflict across different levels of ethnic composition.
However, when accounting for ethnic fractionalization, the coefficient on the trade index in low nightlight areas remains negative and stastically significant, consistent with the baseline results in Table 4.1.

4.2.4 *Water avaibility*

We next investigate whether the pacifying effect of trade exposure varies with local resource endowments. Water availability, derived from the FAO–GAEZ database (FAO & IIASA, 2020), serves as a proxy for agricultural potential and economic activity, given its close link to crop productivity and the prominence of vegetables trade in Nigerian ports.

The intuition is that in ADM2 units with lower water availability, typically more fragile and rural, the trade index should reduce conflict by stabilizing consumption and providing alternative income channels. This mechanism is consistent with evidence that economic integration can mitigate consumption related conflicts by improving access to markets and smoothing shocks (Amodio et al., 2024).

However, the empirical results in Table 4.3 reveal a more complex pattern. Contrary to expectations, the effect of trade exposure turns positive in areas with higher and low water availability, where agricultural and economic activities are already developed. This suggests that, rather than amplifying stability, trade may in resourch rich area vould reinforce existing inequalities or intensify competition over profitable markets (Berman et al., 2017; Michalopoulos & Papaioannou, 2013).

Table 4.3 Water Availability: Levels & Heterogeneity

	Terciles
Index × Water tercile: Low	6.48e-08
(1 = below 33%)	(8.76e-08)
Index × Water tercile: Medium	-2.597e-07**
(1 = between 33% - 66%)	(8.75e-08)
Index × Water tercile: High	3.78e-08
(1 = above 66%)	(9.38e-08)
Nightights intensity	-0.834***
	(0.239)
Observations	1620
Time Fixed effect	YES
District Fixed Effect	YES
Adj. R^2	0.669

Water availability is the ADM2 area-weighted mean of a FAO raster, then split into terciles (cut at the 33rd/66th percentiles: Low/Medium/High). Because levels are absorbed by ADM2 fixed effects, we report Index slopes by water group (interactions). All models include year & ADM2 FE with SE clustered at ADM2. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Overall, the findings indicate that trade openness exerts a stabilizing influence mainly in less developed and resource constrained local economies, a result in line with broader evidence on the heterogeneous development impacts of market integration (Amodio et al., 2024; Dix-Carneiro & Kovak, 2023).

Chapter 5

Conclusion

This thesis has explored how exposure to international maritime trade shapes the likelihood of conflict across Nigerian regions. The analysis combined information on port level activity, maritime connectivity, and geography to build an index of local exposure to global trade shocks. The evidence points to a heterogeneous pattern: the effect of trade on conflict varies widely across space, sectors, and local resource conditions.

5.1 Main findings

Baseline and falsification. For commercial ports, the aggregate effect of the exposure index is small on average, but analysing the heterogeneity of the sample given the development of each ADM2 (proxied by nightlights), we discovered that in relatively less developed areas, higher exposure to trade is associated with lower conflict, while effects in more developed areas are near zero. When index is studied including only ports without commercial relevance, coefficients become weaker and not statistically significant. This falsification supports the interpretation that the mechanism runs through international trade, in particolar access to international market, rather than mere proximity to coastal infrastructure.

Geographic heterogeneity. We studied different heterogeneity to study how the relationship varies across different dimension. Results show that there is a stabilizing effect stronger in the eastern belt than in the west. This spatial asymmetry matches economic patterns: eastern ADM2s are more rural, as shown by the proxy of nightlights, making them more sensitive to trade induced welfare improvements.

Sector and trade flow heterogeneity. Result shows that the negative association is more pronounced for ports oriented toward the one that trade vegetables, as second most traded industry, and imports more with respect of what they export. This is consistent with our theory of consumption smoothing channel.

Tribes. Taking into account for tribal domanince and fractorization, proxied by ethnicity. We saw that the coefficients are positive but not statistically significant.

Resource endowments (water). Interactions with water availability (a proxy for agricultural potential) indicate that the pacifying effect concentrates in resource constrained areas. Where water availability is higher, hence greater agricultural rents or competition, the relationship can weaken or even turn positive, suggesting that trade may intensify distributional tensions when valuable rents are at stake.

5.2 Interpretation

Overall, the findings support that consuption related mechanism are more directly influences by trade exogenous shock, especially in poorer ADM2.
In contrast, where economic rents and pre existing advantages are larger, trade and access to ports may lose significane and in some case can show signals that could have a destabilizing effect. In short, trade is stabilising where markets fill and income opportunities, while it risks destabilising where it amplifies returns and contestable surpluses.

5.3 Limitations

While the findings are robust across specifications reported here, several caution should be taking in consideration.

First measurement. CPT and LSCI are new port indicators available for a shorter window, for which we built to the study period and the other port indicators, sectoral and flow classifications rely on IMF PortWatch assumption available from 2019, future work should verify the validity of those assumption pre 2019. Nightlights and water availability are proxies for income and agricultural potential rather than direct measures.
Second, we used inverse distance to the nearest port aiming to abstract from the actual situation of infrastructures. In this way we didn't take into account for road connections and trasports nework, future work should embed explicit transport networks and travel times.
Third conflict counts aggregate heterogeneous events; although the interpretation emphasizes consumption related mechanisms, misclassification remains possible.
Lastely, the design mitigates simultaneity by using lagged port shocks and fixed effects, and the falsification test helps; still, unobserved time varying shocks correlated with both exposure and conflict and the risk of reverse causality cannot be fully ruled out.

5.4 Policy implications

The heterogeneity of effects observed across regions suggests that how a country integrates its different regions and sector into global markets is as important as the extent of its trade openness. Policies should focus on the qualitative dimensions of integration such us inclusivenes and institutional capacity.

In particular, the findings indicate that trade infrastructure and redistributive policies are crucial to prevent the concentration of gains in port proximate areas, because where trade activity is most intense, improving local governance and transparency is an important channel that can be used to mitigate distributional tensions, increasing the opportunity cost of violence.
Conversely, in inland regions an effective policy should be reducing effective distance to ports investing in infrastructure. In this way, the stabilizing effect of trade exposure could be amplifued by making accessible the global market and adding more economic diversification.
Finally, a crucial policy is the intervention in more developed or areas with more water availability, in which trade expansion should be accompanied by institutional mechanisms that prevent rapacity based conflict.

5.5 External validity and future research

Although Nigeria's coastal geography and regional fragmentation make it a compelling test case, external validity should be assessed cautiously.
Extending the approach to other coastal African economies and to land linked settings would clarify the role of transport networks.
In particular, four concrete extensions emerge: (i) extention to all Nigeria ADM2, and not only to the coastal ones. (ii) Network based exposure, allowing exposure to multiple ports and it could be added the possibility to weight the port based on their economic relevance. (iii) Micro outcomes such as link exposure to prices and employment to pin down the welfare more directly. (iv) Institutional moderators: interact exposure with local governance quality, education and curruption.

5.6 Contribution

Empirically, the thesis aim to contribute to the literatiure introducing a geography exposure indec that link port level shocks and connectivity to local conflict dynamics, accounting for heterogeneity by development, location, sector, tribes and resource endowment.
It aim to ptove a subnational evaluation of trade to shock moving beyond national averages. Results suggest that trade could be a pacifing channel, but not everywhere and not unconditionally. Thus, trade openness contributes to stability only when broadens economic opportunities as its gains are inclusive and not captured by narrow interests, while integration that amplifies rent competition can sustain or even intensify conflict. Effective governance must therefore address this conditional channel. Recognizing this structural asymmetry is essential for policy design.

Appendix A

Conflict

This appendix provides additional information on the conflict data and classification procedures employed in the main analysis

We focused on consumption related conflict, filtering the dataset ACLED (2025) to include only events not associated with predatory behaviour.
This distinction is important because rapacity related events generally reflect violent and opportunistic actions, whereas non rapacity conflicts are more closely linked to income related grievances and economic distress.

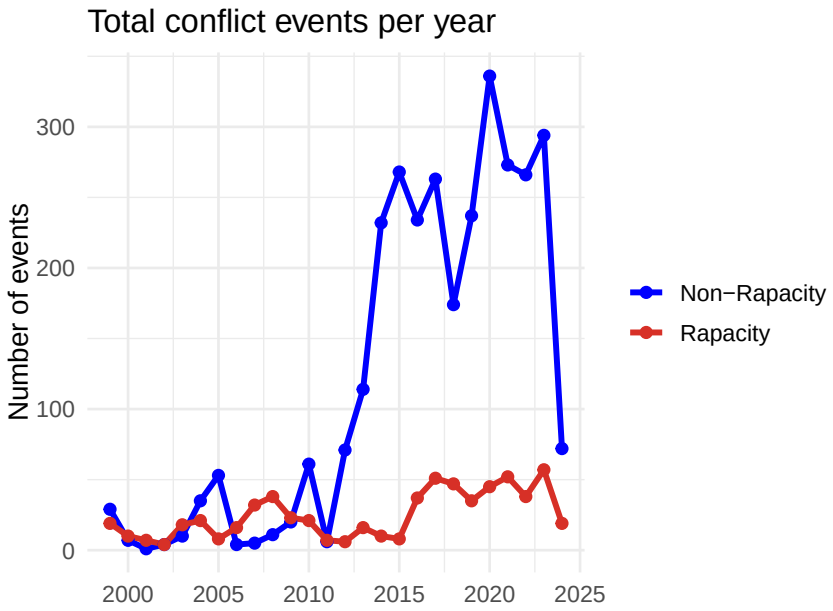


Figure A.1 Dominant conflict type

Figure A.1 illustrates the temporal evolution of conflict events, showing the total number of incidents of each type per year across ADM2 units.
Non rapacity conflicts increase sharply after 2015, consistent with the patterns identified in the main analysis, while rapacity events remain relatively stable over

time.
This divergence suggests that the two categories may be driven, at least in part, by different underlying mechanisms.

Table A.1 Panel coverage and sparsity of conflict events

Conflict type	Average	SD	Peak activity
Non-rapacity	0.66	2.40	336 events in 2020
Rapacity	0.14	0.61	57 events in 2023

Table A.1 shows that, on average, each ADM2 experiences almost no conflict events per year.
However, the high standard deviation of non-rapacity events reveals substantial heterogeneity across regions, with a few ADM2s concentrating most of the violence. This sparsity and imbalance justify the inclusion of ADM2 fixed effects in the empirical model.

Appendix B

Nightlights

This appendix provides additional context on the control variable, nightlights.

In the Data chapter we noted visible discontinuities in the series, which Li et al. (2020) attribute to satellite technology changes and sensor fluctuations.
To rule out other drivers, we compared national trends in nightlights with two plausibly related aggregates: conflict intensity (combining rapacity and consumption-related events) and oil–gas rents, given the prominence of minerals and oil in Nigeria's trade (International Monetary Fund & University of Maryland, 2025).

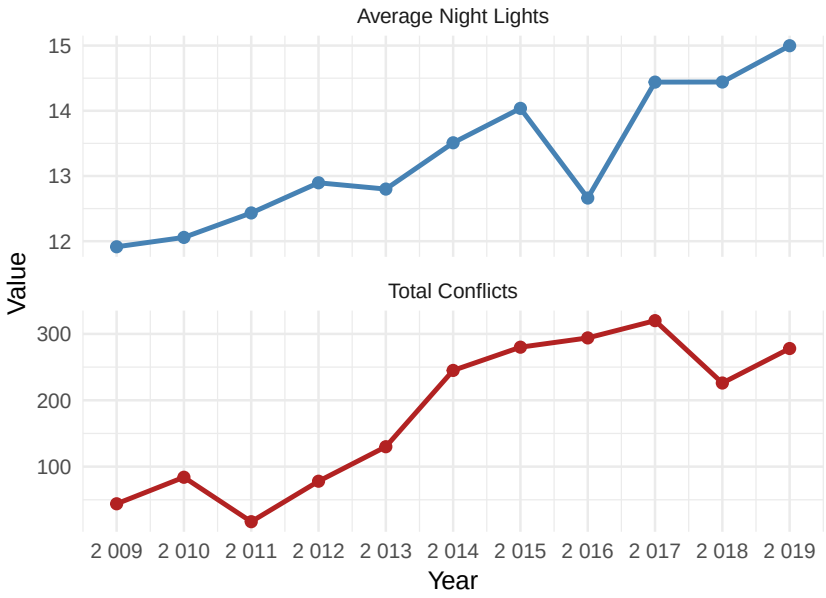


Figure B.1 Conflict and nightlight

Figure B.1 shows the patterns are consistent with dual influences. In 2013 and especially 2016, declines in nightlights coincide with marked increases in conflict, suggesting real economic disruption alongside potential measurement noise.

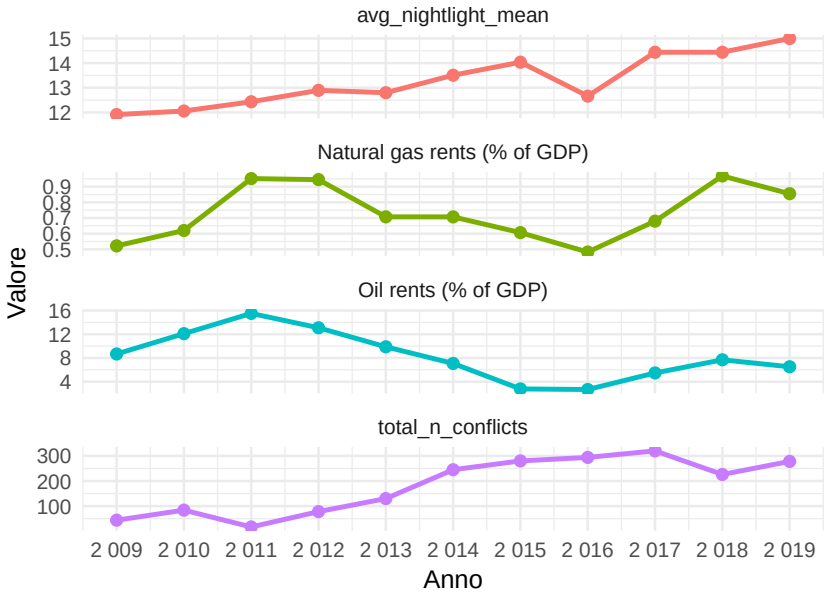


Figure B.2 Confroning conflict, nightlight and different type of rent

Figure B.2 shows a similar pattern and, in particolar, that the 2016 episode also aligns with a sharp fall in traded-goods rents Those observations linked with the sensor moviment could explain the drop Of course, part of the discontinuity in nightlights reflects known sensor and constellation changes Li et al. (2020). Yet the comovement we observe, nightlights falling in 2013 and especially 2016 while conflict rises and traded goods rents contract, suggests a substantive pattern beyond pure measurement noise. This alignment points to plausible real-economy disruption coinciding with the technical break. While we do not claim causality here, the repeated timing and direction of these series indicate a relationship that merits deeper investigation rather than being dismissed as a data artifact alone.

Appendix C

Heterogeneity

This appendix provides additional information about the measurement used as heterogeneity employed used in the empirical strategy.
In our analyses we accounted for different spatial heterogeneity, in particular for geographic, sectorial, social and environmental one.

Starting from geographical dimension, we studied both the differences between East and Ovest and the ones between Nord and South. We costructed the dummies using respectively the longitude and latitude median. Nord and South, as shown in table C.1, display strong differences in nightlight concentrations, a proxy for economic activity: the South is brighter on average and more dispersed, while the North is dimmer and more stable. This suggests that in southern regions, the wealthiest areas are wealthier than those in the north.

Table C.1 Summary statistics by North/South macro-region

Macro-region	n	Mean NTL	SD NTL	Mean trade	SD trade	Mean conflicts	SD conflicts
North	748	1.35	0.885	9.07	50.0	0.707	1.89
South	1232	2.59	0.932	131551.0	2782390.0	1.19	3.55

Notes: NTL = log night-time lights. Trade = *indice2_main*. Conflicts = number of consumption-related events.

However, the corresponding regressions are not statistically significant, so we do not rely on the North/South split in the main analysis.

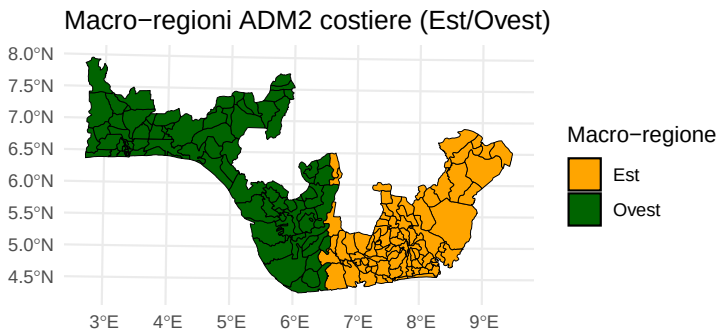


Figure C.1 est and ovest

We therefore focus on the East West division. Figure Figure C.1 shows the spatial partition and, following the intuition of the Panel B of Table C.1, we analysed the logarithmic distribution of nightlights by macro region Figure C.2.

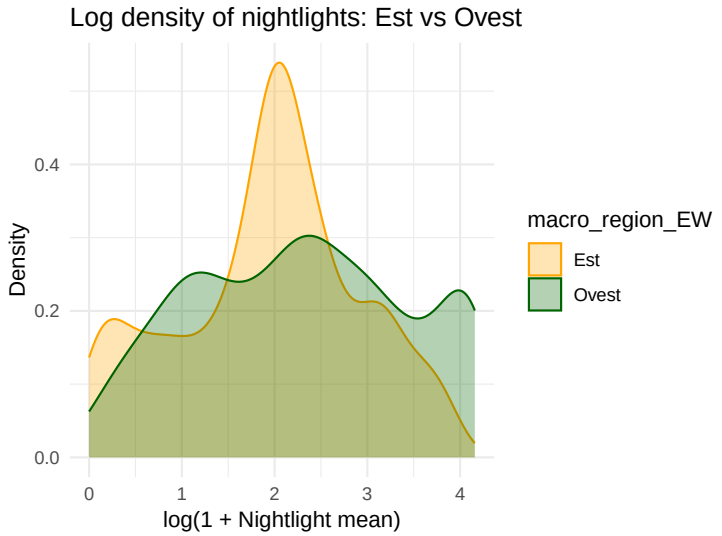


Figure C.2 log density of nightlights

The pattern confirms our intuition: the East, in orange, concentrates around 2 with more mass at lower values, while the West, in green, features a heavier right tail up to 4. This suggest that the western ADM2 units are, on average, brighter and more dispersed.

A noteworthy observation is the almost perfect symmetry between the geographic (East/West) and the sectoral dimensions (Import/Export and Vegetables-

first/Chemicals-first): they cover largely the same ADM2 units, which strengthens the interpretation of our heterogeneity results. This symmetries are shown in Figure C.1, C.3 and C.4

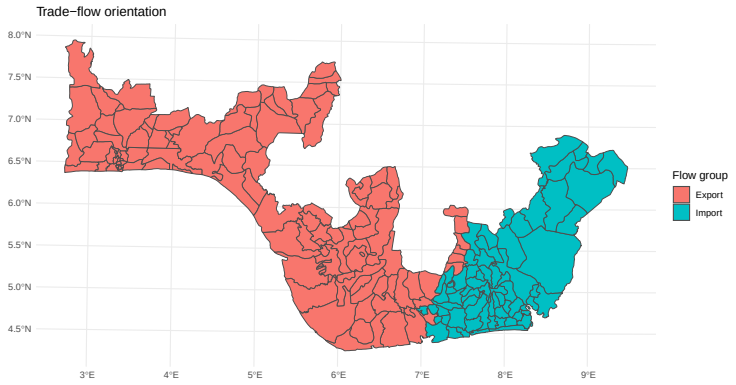


Figure C.3 Import and export

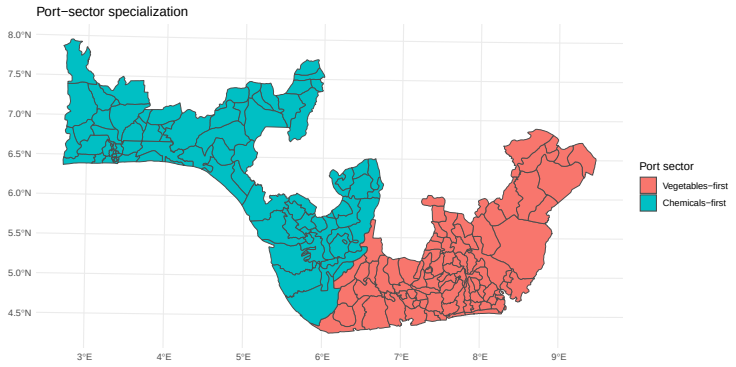


Figure C.4 Vegetable and chemicals

Appendix D

Replication Code

This appendix provides the R code used to construct the analysis

Listing D.1 Full pipeline

```
remove(list = ls())

library(terra)
library(exactextractr)
library(sf)
library(dplyr)
library(ggplot2)
library(ggrepel)
library(viridis)
library(haven)
library(fixest)
library(modelsummary)
library(labelled)
library(SPEI)
library(conleyreg)
library(raster)
library(readxl)
library(ncdf4)
library("RNetCDF")
library(patchwork)
library(sf)
library(fixest)

#step 1: shapefile
shp <- read_sf(paste0(getwd(), "/TM_WORLD_BORDERS-0.3.shp"))
shp <- st_make_valid(shp)
shp <- shp %>%
  mutate(id = 1:n(),
         NAME = tolower(NAME))

# Load names of Subsaharan countries and select them in shp
load("afr_names.rda")
sub_shp <- subset(shp, ISO3 %in% afr_names$iso3c)
# Subset the sub_shp object to include only Nigeria (ISO3 code "NGA")
shp <- shp %>% filter(NAME %in% "nigeria")

#admin2 level
nig <- read_sf(paste0(getwd(), "/nga_polnda_adm2_1m_salb.shp"))
```

```
shp <- shp %>%
  st_transform("+init=EPSG:26393")

nig <- nig %>%
  st_transform("+init=EPSG:26393")

#creating a grid
grid <- st_make_grid(nig, cellsize = 100000) # takes shp and returns a grid

# We need to remove the part of squares not perfectly overlapping with shp
grid <- grid %>% st_intersection(nig) # overlapping squares, missing values

# Compute the area of cells
area <- st_area(grid)
area <- as.numeric(area); area
to_rm <- which(area <= 7e+08); to_rm
# Drop smallest cells
grid <- grid[-to_rm]

# In order to match the grid with other shapefiles, we need the grid to be a shapefile
grid <- st_as_sf(grid)
# We can also assign an id to each cell
grid <- grid %>%
  mutate(FIP = 1:n())
# Adjust path to point to the unzipped .shp file
shapefile_path <- "ports.shp"

# Read shapefile
ports_data <- st_read(shapefile_path)

ports_data <- st_transform(ports_data, "+init=EPSG:26393")
ports_data <- st_transform(ports_data, st_crs(nig))

# Plot
ggplot() +
  geom_sf(data = nig, fill = "gray90", color = "black") +  # Admin2 map
  geom_sf(data = grid, fill = NA, color = "black", linewidth = 0.2) +  # Grid overlay
  geom_sf(data = ports_data, color = "red", size = 2) +  # Ports as red dots
  theme_minimal() +
  ggtitle("Nigerian Ports with Admin2 Boundaries and Grid")
#add the Onne port, absent in the dataset
cols_no_geom <- setdiff(names(ports_data), "geometry")
nuovo_porto <- as.data.frame(matrix(NA, nrow = 1, ncol = length(cols_no_geom)))
names(nuovo_porto) <- cols_no_geom
nuovo_porto$MAIN_PORT_ <- "ONNE"
nuovo_porto$LNG_DD <- 7.0361
nuovo_porto$LAT_DD <- 4.7444

nuovo_porto_sf <- st_as_sf(
```

```
88    nuovo_porto,
89    coords = c("LNG_DD", "LAT_DD"),
90    crs = 4326,
91    remove = FALSE
92  )
93  st_geometry(nuovo_porto_sf) <- st_sfc(st_point(c(7.0361, 4.7444)),
        crs = 4326)
94  nuovo_porto_sf <- st_transform(nuovo_porto_sf, st_crs(ports_data))
95
96  #Avoid duplicates
97  if ("MAIN_PORT_" %in% names(ports_data)) {
98    ports_data <- ports_data %>% filter(is.na(MAIN_PORT_) | MAIN_PORT_
        != "ONNE")
99  }
100
101 add_missing_cols <- function(x, cols) {
102   missing <- setdiff(cols, names(x))
103   if (length(missing) > 0) {
104     for (m in missing) x[[m]] <- NA
105   }
106   x[, cols]
107 }
108 common_cols <- union(names(ports_data), names(nuovo_porto_sf))
109 ports_data_aligned     <- add_missing_cols(ports_data, common_cols)
110 nuovo_porto_sf_aligned <- add_missing_cols(nuovo_porto_sf, common_
      cols)
111
112 ports_data <- rbind(ports_data_aligned, nuovo_porto_sf_aligned)
113 #plot again
114 ggplot() +
115   geom_sf(data = nig, fill = "gray90", color = "black") +  # Admin2
        map
116   geom_sf(data = grid, fill = NA, color = "black", linewidth = 0.2) +
          # Grid overlay
117   geom_sf(data = ports_data, color = "red", size = 2) +  # Ports as
        red dots
118   theme_minimal() +
119   ggtitle("Nigerian Ports with Admin2 Boundaries and Grid")
120 nigeria_boundary <- st_boundary(st_union(nig))
121
122 #Coastal ADMIN
123 coastal_ADM1_NAME <- c(
124   "Lagos", "Ogun", "Ondo", "Delta", "Bayelsa", "Rivers", "Akwa Ibom",
        "Cross River", "Abia"
125 )
126
127 coastal_admin1 <- nig %>%
128   filter(ADM1_NAME %in% coastal_ADM1_NAME)
129 admin2_in_coastal_admin1 <- st_join(nig, coastal_admin1, join = st_
      within, left = FALSE)
130
131 #merge
132 admin2_in_coastal_admin1 <- st_join(nig, coastal_admin1, join = st_
      within, left = FALSE)
133 coastal_admin2_master <- admin2_in_coastal_admin1
134
135 #Cleaning
```

```
136 if ("ADM2_NAME.x" %in% names(coastal_admin2_master)) {
137   coastal_admin2_master <- coastal_admin2_master %>% dplyr::rename(
          ADM2_NAME = ADM2_NAME.x)
138 }
139
140 if ("ADM2_CODE.x" %in% names(coastal_admin2_master)) {
141   coastal_admin2_master <- coastal_admin2_master %>% dplyr::rename(
          ADM2_CODE = ADM2_CODE.x)
142 }
143
144 admin2_in_coastal_admin1 <- coastal_admin2_master
145
146 #buffer
147 convert_to_decimal <- function(deg, min, dec, hemi) {
148   coord <- deg + min/60 + dec/3600
149   coord <- ifelse(hemi %in% c("S", "W"), -coord, coord)
150   return(coord)
151 }
152
153 ports_data <- ports_data %>%
154   mutate(
155     LAT_dec = convert_to_decimal(LAT_DEG, LAT_MIN, LAT_DD, LAT_HEMI),
156     LON_dec = convert_to_decimal(LONG_DEG, LONG_MIN, LNG_DD, LONG_
            HEMI)
157   )
158
159 #sf object
160 ports_sf <- st_as_sf(ports_data, coords = c("LON_dec", "LAT_dec"),
      crs = 4326)
161 ports_proj <- st_transform(ports_sf, 32631)
162 ports_buffer <- st_buffer(ports_proj, dist = 1000000)  # buffer di
      1000km
163 ports_buffer <- st_make_valid(ports_buffer)
164 admin2_proj <- st_transform(admin2_in_coastal_admin1, st_crs(ports_
      buffer))
165 coastal_buffer_admin2 <- admin2_proj %>%
166   filter(st_intersects(admin2_proj, ports_buffer, sparse = FALSE)
        [,1])
167
168 #minimum distance
169 dist_ports <- st_distance(coastal_buffer_admin2, ports_proj)
170 coastal_buffer_admin2$dist_min_port <- apply(as.matrix(dist_ports),
      1, min)/1000
171
172 ####CONFLICT####
173 conflict <- read_excel("Cartel2.xlsx")
174 rapacity_sub_events <- c(
175   "Looting/property destruction",
176   "Abduction/forced disappearance",
177   "Armed clash",
178   "Attack",
179   "Shelling/artillery/missile attack",
180   "Sexual violence"
181 )
182
183
184 non_rapacity_sub_events <- c(
```

```
185   "Peaceful protest",
186   "Violent demonstration",
187   "Protest with intervention",
188   "Agreement",
189   "Change to group/activity",
190   "Disrupted weapons use",
191   "Arrests",
192   "Headquarters or base established",
193   "Non-violent transfer of territory",
194   "Non-state actor overtakes territory",
195   "Government regains territory",
196   "Other"
197 )
198
199 conflict_non_rapacity <- conflict %>%
200   filter(
201     sub_event_type %in% non_rapacity_sub_events |
202       grepl("loot|steal|raid|resource|gold|oil|land|plunder|livestock
              |pasture|farm|grazing",
203             tolower(notes))
204   )
205
206
207 conflict_non_rapacity_clean <- conflict_non_rapacity[complete.cases(
      conflict_non_rapacity$longitude, conflict_non_rapacity$latitude),
      ]
208
209 # Count the number of removed lines
210 rows_removed <- nrow(conflict_non_rapacity) - nrow(conflict_non_
      rapacity_clean)
211 conflict_sf <- st_as_sf(conflict_non_rapacity_clean, coords = c("
      longitude", "latitude"), crs = st_crs(nig))
212
213 # Ensure both are in the same CRS
214 grid_transformed <- st_transform(grid, "+init=EPSG:26393")
215 conflict_sf_transformed <- st_transform(conflict_sf, "+init=EPSG
      :26393")
216 conflict_sf_transformed <- st_transform(conflict_sf, crs = 26393)
217
218 # Step 2: Re-transform conflict_sf to EPSG:26393
219 conflict_sf_transformed <- st_transform(conflict_sf, crs = 26393)
220
221 # Step 3: Perform spatial join and summarize
222 conflict_sf1 <- st_join(grid_transformed, conflict_sf_transformed)
      %>%
223   group_by(FIP, admin2, year) %>%
224   summarise(n_conflicts = n(), .groups = 'drop')
225 conflict_sf_transformed1 <- conflict_sf_transformed %>%
226   mutate(conflict_id = row_number())
227
228 conflict_sf3 <- st_join(grid_transformed, conflict_sf_transformed)
      %>%
229   group_by(FIP, year, event_type,fatalities,event_id_cnty) %>%
230   summarise(n_conflicts = n())
231 st_crs(conflict_sf) <- 4326
232 conflict_sf <- st_transform(conflict_sf, st_crs(nig))
233
```

```
coastal_admin2 <- st_transform(admin2_in_coastal_admin1, crs = st_crs(conflict_sf))
within_matrix <- st_within(conflict_sf, coastal_admin2, sparse = FALSE)
#VERIFY
inside_any <- apply(within_matrix, 1, any)
#filter
conflict_coastal <- conflict_sf[inside_any, ]

# Perform the spatial join to get conflicts within the master coastal ADM2s
coastal_admin2_master_proj <- st_transform(coastal_admin2_master, crs = 4326)
coastal_admin2_master_proj <- st_transform(coastal_admin2_master, crs = 26393)
st_crs(conflict_sf_transformed) <- 4326
conflict_sf_transformed <- st_transform(conflict_sf_transformed, 4326)
conflict_sf_transformed <- st_transform(conflict_sf_transformed, 26393)
# Using st_intersects for points within polygons, or st_within for stricter containment
conflict_coastal <- st_join(conflict_sf_transformed, coastal_admin2_master_proj, join = st_intersects, left = FALSE)

conflict_sf <- sf::st_transform(conflict_sf, sf::st_crs(coastal_admin2_master))

##Join spaziale
ev_join <- sf::st_join(conflict_sf, coastal_admin2_master, join = sf::st_within, left = FALSE)
ev_df   <- as.data.frame(sf::st_drop_geometry(ev_join))

adm2_vec  <- unique(as.character(as.data.frame(sf::st_drop_geometry(coastal_admin2_master))$ADM2_NAME))
years_vec <- sort(unique(ev_df$year))   # oppure: years_vec <- seq(2009, 2019)

counts_raw <- aggregate(list(n_conflicts = ev_df$year),
                        by = list(ADM2_NAME = ev_df$ADM2_NAME,
                                  year      = ev_df$year),
                        FUN = length)
#panel
panel <- merge(data.frame(ADM2_NAME = adm2_vec, stringsAsFactors = FALSE),
               data.frame(year = years_vec), all = TRUE)
conflicts_adm2_year <- merge(panel, counts_raw, by = c("ADM2_NAME","year"), all.x = TRUE)
conflicts_adm2_year$n_conflicts[is.na(conflicts_adm2_year$n_conflicts)] <- 0L

###NIGHTLIGHTS####
dir.create("nightlights_data", showWarnings = FALSE)
nig <- st_transform(nig, crs = "EPSG:4326")
#list
raster_files <- list.files("nightlights/", pattern = "\\.tif$", full.names = TRUE)
```

```
274 r <- rast("nightlights/Harmonized_DN_NTL_2009_calDMSP.tif")
275 nig <- st_transform(nig, crs = crs(r))
276
277 nig_vect <- vect(nig)
278 r_crop <- crop(r, nig_vect)
279 r_mask <- mask(r_crop, nig_vect)
280
281 #output folder
282 dir.create("nightlights_nigeria", showWarnings = FALSE)
283
284 # 5. Loop for each raster
285 for (file in raster_files) {
286   r <- rast(file)
287   r_crop <- crop(r, vect(nig))
288   r_mask <- mask(r_crop, vect(nig))
289   plot(r_mask)
290   year <- gsub("[^0-9]", "", basename(file))
291   out_name <- paste0("nightlights_nigeria/nightlight_nigeria_", year,
          ".tif")
292
293   writeRaster(r_mask, out_name, overwrite = TRUE)
294   cat("Salvato:", out_name, "\n")
295 }
296
297 coastal_admin2 <- st_transform(coastal_admin2, crs = "EPSG:4326")
298 coastal_vect <- vect(coastal_admin2)
299
300 # list
301 raster_files <- list.files("nightlights/", pattern = "\\.tif$", full.
      names = TRUE)
302
303 # output for coastal
304 dir.create("nightlights_coastal", showWarnings = FALSE)
305
306 for (file in raster_files) {
307   r <- rast(file)
308   coastal_vect_proj <- project(coastal_vect, crs(r))
309   r_crop <- crop(r, coastal_vect_proj)
310   r_mask <- mask(r_crop, coastal_vect_proj)
311   plot(r_mask)
312   year <- gsub("[^0-9]", "", basename(file))
313   out_name <- paste0("nightlights_coastal/nightlight_coastal_", year,
          ".tif")
314   writeRaster(r_mask, out_name, overwrite = TRUE)
315   cat("Salvato:", out_name, "\n")
316 }
317
318
319 nightlight_stats <- terra::extract(r_mask, vect(admin2_in_coastal_
      admin1), fun=mean, na.rm=TRUE)
320
321 # add admin2 coastal
322 admin2_in_coastal_admin1$nightlight_mean <- nightlight_stats[,2]
323 admin2_in_coastal_admin1$nightlight_norm <- (admin2_in_coastal_admin1
      $nightlight_mean - min(admin2_in_coastal_admin1$nightlight_mean,
      na.rm=TRUE)) /
324   (max(admin2_in_coastal_admin1$nightlight_mean, na.rm=TRUE) - min(
```

```
      admin2_in_coastal_admin1$nightlight_mean, na.rm=TRUE))
325
326 ####SHOCK Container Port Throughput
327 ContPortThroughput<- read_excel("US.ContPortThroughput_20250723_
      173714.xlsx")
328 ContPortThroughput <- as.data.table(ContPortThroughput)
329 ContPortThroughput <- ContPortThroughput %>%
330   filter(Economy_Label == "Nigeria")
331
332 cols_teu <- grep("^20(0[0-9]|1[0-9])_TEU_Twenty_foot_Equivalent_Unit_
      Value$", names(ContPortThroughput), value = TRUE)
333 cols_to_keep <- c("Economy_Label", cols_teu)
334 ContPortThroughput <- ContPortThroughput[, ..cols_to_keep]
335 ContPortThroughput[, (cols_teu) := lapply(.SD, function(x) as.numeric
      (as.character(x))), .SDcols = cols_teu]
336 ContPortThroughput_long <- melt(
337   ContPortThroughput,
338   id.vars = "Economy_Label",
339   measure.vars = patterns("^20(0[0-9]|1[0-9])_TEU_Twenty_foot_
      Equivalent_Unit_Value$"),
340   variable.name = "year",
341   value.name = "teu"
342 )
343
344 ContPortThroughput_long[, year := as.integer(gsub("_.*", "", names(
      ContPortThroughput)[
345   grep("^20(0[0-9]|1[0-9])_TEU_Twenty_foot_Equivalent_Unit_Value$",
      names(ContPortThroughput))
346 ][year]))]
347
348 setorder(ContPortThroughput_long, year)
349
350 ContPortThroughput_long[, shock := c(NA, diff(log(teu)))]
351 ####LSCI#####
352 LSCI<- read_excel("LSCI.xlsx")
353 LSCI <- as.data.table(LSCI)
354 LSCI <- LSCI %>%
355   filter(Economy_Label == "Nigeria")
356
357
358 cols_quarterly <- names(LSCI)[-1]
359 LSCI_long <- melt(LSCI, measure.vars = cols_quarterly,
360                   variable.name = "Quarter", value.name = "LSCI_value
                      ")
361
362 #cleaning
363 parse_quarter_date <- function(colname) {
364   date_part <- sub("_Index.*", "", colname)
365   date_part <- gsub("\\s+", " ", date_part)
366   date_part <- gsub("\\.", "", date_part)
367   parts <- strsplit(date_part, " ")[[1]]
368   month_str <- parts[1]
369   year_str <- parts[2]
370   month_map <- c(
371     Jan = 1, Feb = 2, Mar = 3, Apr = 4,
372     May = 5, Jun = 6, Jul = 7, Aug = 8,
373     Sep = 9, Oct = 10, Nov = 11, Dec = 12
```

```
  )
  if (!month_str %in% names(month_map)) return(NA)
  month_num <- month_map[[month_str]]
  date_str <- sprintf("%s-%02d-15", year_str, month_num)
  as.Date(date_str)
}

dates <- sapply(cols_quarterly, parse_quarter_date)

LSCI_long <- melt(LSCI,
                  id.vars = "Economy_Label",
                  measure.vars = cols_quarterly,
                  variable.name = "quarter_label",
                  value.name = "LSCI_value")
LSCI_long[, date := as.Date(dates[quarter_label])]

LSCI_long[, year := year(date)]
LSCI_long[, quarter := quarter(date)]
LSCI_long[, LSCI_value := as.numeric(LSCI_value)]  # assicura che sia numerico
LSCI_annual <- LSCI_long[, .(LSCI_annual = mean(LSCI_value, na.rm = TRUE)), by = year]

###Constructing the index####
nightlight_list <- list()

for (file in raster_files) {
  r <- rast(file)
  year <- as.integer(gsub("[^0-9]", "", basename(file)))

  nightlight_vals <- terra::extract(r, vect(admin2_in_coastal_admin1), fun=mean, na.rm=TRUE)[,2]

  tmp_dt <- data.table(
    ADM2_CODE = admin2_in_coastal_admin1$ADM2_CODE,
    ADM2_NAME = admin2_in_coastal_admin1$ADM2_NAME,
    year = year,
    nightlight_mean = nightlight_vals
  )

  nightlight_list[[as.character(year)]] <- tmp_dt
}

nightlight_dt <- rbindlist(nightlight_list)

#first merge
coastal_buffer_admin2 <- merge(nightlight_dt, coastal_buffer_admin2 , by = "ADM2_CODE", all.x = TRUE)
coastal_buffer_admin2[, ADM2_NAME.y := NULL]
final_df <- merge(coastal_buffer_admin2, LSCI_annual, by = "year", all.x = TRUE)

#normalize distance
final_df$export_proxy <- 1 / (final_df$dist_min_port + 1)

#skeleton of panel data
years <- unique(LSCI_annual$year)
```

```
adm2s <- unique(coastal_admin2_master$ADM2_NAME)
panel_skeleton <- expand.grid(ADM2_NAME = adm2s, year = years)
print(paste("Scheletro creato. Numero di righe:", nrow(panel_skeleton
    )))
final_regression_data <- merge(panel_skeleton, LSCI_annual, by = "
    year", all.x = TRUE)
print(paste("Dopo unione con LSCI_annual. Numero di righe:", nrow(
    final_regression_data)))
final_regression_data <- merge(final_regression_data,
    ContPortThroughput_long[, c("year", "shock")], by = "year", all.x
    = TRUE)
print(paste("Dopo unione con ContPortThroughput_long. Numero di righe
    :", nrow(final_regression_data)))
final_regression_data <- merge(final_regression_data, nightlight_dt,
    by = c("ADM2_NAME", "year"), all.x = TRUE)
print(paste("Dopo unione con nightlight_annual. Numero di righe:",
    nrow(final_regression_data)))
final_regression_data <- merge(final_regression_data, conflicts_
    aggregated, by = c("ADM2_NAME", "year"), all.x = TRUE)
print(paste("Dopo unione con conflicts_aggregated. Numero di righe:",
     nrow(final_regression_data)))
setDT(final_df)
norm_vals <- final_df |>
  dplyr::select(ADM2_NAME, dist_min_norm, dist_min_port) |>
  dplyr::distinct()
norm_vals[, names(norm_vals)[grepl("\\.y$", names(norm_vals))] :=
    NULL]
x_cols <- names(norm_vals)[grepl("\\.x$", names(norm_vals))]
setnames(norm_vals, old = x_cols, new = gsub("\\.x$", "", x_cols))
# adding coastal buffer admin2

setDT(coastal_buffer_admin2)
coastal_buffer_admin2[, names(coastal_buffer_admin2)[grepl("\\.y$",
    names(coastal_buffer_admin2))] := NULL]

coastal_buffer_admin2 <- coastal_buffer_admin2 |>
  left_join(norm_vals, by = "ADM2_NAME")
x_cols <- names(coastal_buffer_admin2)[grepl("\\.x$", names(coastal_
    buffer_admin2))]
setnames(coastal_buffer_admin2, old = x_cols, new = gsub("\\.x$", "",
     x_cols))
dist_one <- as.data.table(st_drop_geometry(coastal_buffer_admin2))[
  , .(dist_min_port = min(as.numeric(dist_min_port), na.rm = TRUE),
      dist_min_norm = first(dist_min_norm)), by = ADM2_NAME]

final_regression_data <- dist_one[final_regression_data, on = "ADM2_
    NAME"]
print(paste("Dopo unione con i dati geografici. Numero di righe:",
    nrow(final_regression_data)))
setDT(final_regression_data)


# adding epsilon to avoid division for 0
epsilon <- 0.000001
final_regression_data[is.na(n_conflicts), n_conflicts := 0]
final_regression_data <- final_regression_data %>%
  mutate(
```

```
467     indice2 = (1 / (dist_min_norm + epsilon)) * shock * lsci_value
468   )
469
470 ###index only for main ports ####
471
472 main_ports <- c("LAGOS", "TIN CAN ISLAND", "PORT HARCOURT",
473                 "ONNE", "WARRI", "CALABAR")
474 #filtering
475 ports_main_sf <- ports_sf %>%
476   filter(MAIN_PORT_ %in% main_ports)
477 ports_main_proj <- st_transform(ports_main_sf, 32631)
478
479
480 # Buffer
481 ports_buffer_main <- st_buffer(ports_main_proj, dist = 1000000)
482 ports_buffer_main <- st_make_valid(ports_buffer_main)
483 admin2_proj_main <- st_transform(admin2_in_coastal_admin1, st_crs(
      ports_buffer_main))
484
485 coastal_buffer_admin2_main <- admin2_proj_main %>%
486   filter(st_intersects(admin2_proj_main, ports_buffer_main, sparse =
        FALSE)[,1])
487
488 dist_ports_main <- st_distance(coastal_buffer_admin2_main, ports_main
      _proj)
489 coastal_buffer_admin2_main$dist_min_port_main <- apply(as.matrix(dist
      _ports_main), 1, min)/1000
490
491 coastal_admin2_master_main <- merge(
492   coastal_admin2_master,
493   st_drop_geometry(coastal_buffer_admin2_main[, c("ADM2_NAME", "dist_
        min_port_main")]),
494   by = "ADM2_NAME",
495   all.x = TRUE
496 )
497
498 #normalizing distances
499 coastal_admin2_master_main <- coastal_admin2_master_main %>%
500   mutate(
501     dist_min_norm_main = (dist_min_port_main - min(dist_min_port_main
          , na.rm = TRUE)) /
502       (max(dist_min_port_main, na.rm = TRUE) - min(dist_min_port_main
            , na.rm = TRUE))
503   )
504
505 coastal_buffer_admin2_main <- merge(
506   nightlight_dt,
507   coastal_buffer_admin2_main,
508   by = "ADM2_CODE",
509   all.x = TRUE
510 )
511 coastal_buffer_admin2_main[, ADM2_NAME.y := NULL]
512
513 final_df_main <- merge(coastal_buffer_admin2_main, LSCI_annual, by =
      "year", all.x = TRUE)
514
515 final_df_main$export_proxy_main <- 1 / (final_df_main$dist_min_port_
```

```
    main + 1)
516
517 final_df_main <- final_df_main %>%
518   mutate(
519     dist_min_norm_main = (dist_min_port_main - min(dist_min_port_main
          , na.rm = TRUE)) /
520       (max(dist_min_port_main, na.rm = TRUE) - min(dist_min_port_main
          , na.rm = TRUE)),
521     export_proxy_main  = ifelse(is.na(export_proxy_main), 0, export_
          proxy_main)
522   )
523 panel_list_main <- list()
524 for (t in LSCI_annual$year) {
525   lsci_t  <- LSCI_annual[year == t, lsci_value][1]
526   shock_t <- ContPortThroughput_long[year == t, shock][1]
527
528   if (is.na(shock_t)) shock_t <- 0
529
530   tmp <- final_df_main %>%
531     mutate(
532       year      = t,
533       shock     = shock_t,
534       Exposure_main = lsci_t * shock_t * dist_min_norm_main,
535       indice2_main  = lsci_t * shock_t * export_proxy_main
536     )
537   panel_list_main[[as.character(t)]] <- tmp
538 }
539
540 panel_data_main <- bind_rows(panel_list_main)
541 setDT(panel_data_main)
542 panel_data_main[, names(panel_data_main)[grepl("\\.y$", names(panel_
    data_main))] := NULL]
543 x_cols_main <- names(panel_data_main)[grepl("\\.x$", names(panel_data
    _main))]
544 setnames(panel_data_main, old = x_cols_main, new = gsub("\\.x$", "",
    x_cols_main))
545
546 years <- unique(LSCI_annual$year)
547 adm2s <- unique(coastal_admin2_master_main$ADM2_NAME)  # <-- usa la
    versione _main
548 panel_skeleton_main <- expand.grid(ADM2_NAME = adm2s, year = years)
549
550 final_regression_data_main <- merge(panel_skeleton_main, LSCI_annual,
     by = "year", all.x = TRUE)
551 final_regression_data_main <- merge(final_regression_data_main,
    ContPortThroughput_long[, c("year", "shock")], by = "year", all.x
    = TRUE)
552 final_regression_data_main <- merge(final_regression_data_main,
    nightlight_dt, by = c("ADM2_NAME", "year"), all.x = TRUE)
553 final_regression_data_main <- merge(final_regression_data_main,
    conflicts_aggregated, by = c("ADM2_NAME", "year"), all.x = TRUE)
554 final_regression_data_main <- merge(final_regression_data_main, st_
    drop_geometry(coastal_admin2_master_main[, c("ADM2_NAME", "dist_
    min_port_main", "dist_min_norm_main")]), by = "ADM2_NAME", all.x =
     TRUE)
555 setDT(final_regression_data_main)
556 final_regression_data_main[is.na(n_conflicts), n_conflicts := 0]
```

```
557
558
559 epsilon <- 1e-6
560 final_regression_data_main <- final_regression_data_main %>%
561   mutate(
562     indice2_main = (1 / (dist_min_norm_main + epsilon)) * shock *
          lsci_value,
563     indice_main  = dist_min_norm_main * shock * lsci_value
564   )
565
566 # creation of index for falsification test: remove the main ports
567 main_ports <- c("LAGOS", "TIN CAN ISLAND", "PORT HARCOURT",
568                 "ONNE", "WARRI", "CALABAR")
569
570 ports_nonmain_sf <- ports_sf %>%
571   filter(!MAIN_PORT_ %in% main_ports)
572
573 #control
574 stopifnot(nrow(ports_nonmain_sf) > 0)
575
576 ports_nonmain_proj <- st_transform(ports_nonmain_sf, 32631)
577 admin2_proj_all    <- st_transform(admin2_in_coastal_admin1, st_crs(
      ports_nonmain_proj))
578 #min distance
579 dist_ports_nonmain <- st_distance(admin2_proj_all, ports_nonmain_proj
      )  # in metri
580 dist_min_port_nonmain_km <- apply(as.matrix(dist_ports_nonmain), 1,
      min) / 1000
581
582 admin2_with_nonmain <- admin2_proj_all %>%
583   mutate(dist_min_port_nonmain = dist_min_port_nonmain_km)
584
585 coastal_admin2_master_nonmain <- merge(
586   coastal_admin2_master,
587   st_drop_geometry(admin2_with_nonmain[, c("ADM2_NAME", "dist_min_
        port_nonmain")]),
588   by = "ADM2_NAME",
589   all.x = TRUE
590 )
591
592 coastal_admin2_master_nonmain <- coastal_admin2_master_nonmain %>%
593   mutate(
594     dist_min_norm_nonmain = (dist_min_port_nonmain - min(dist_min_
          port_nonmain, na.rm = TRUE)) /
595       (max(dist_min_port_nonmain, na.rm = TRUE) - min(dist_min_port_
            nonmain, na.rm = TRUE)),
596     export_proxy_nonmain = 1 / (dist_min_port_nonmain + 1)  # stessa
          logica dell'export_proxy_main
597   )
598
599 #panel
600 final_df_nonmain <- merge(
601   nightlight_dt,
602   st_drop_geometry(coastal_admin2_master_nonmain[, c("ADM2_NAME","
        dist_min_port_nonmain","dist_min_norm_nonmain","export_proxy_
        nonmain")]),
603   by = "ADM2_NAME",
```

```
  all.x = TRUE
)

final_df_nonmain <- merge(final_df_nonmain, LSCI_annual, by = "year",
    all.x = TRUE)

panel_list_nonmain <- list()
for (t in LSCI_annual$year) {
  lsci_t  <- LSCI_annual[year == t, lsci_value][1]
  shock_t <- ContPortThroughput_long[year == t, shock][1]
  if (is.na(shock_t)) shock_t <- 0

  tmp <- final_df_nonmain %>%
    mutate(
      year               = t,
      shock              = shock_t,
      Exposure_nonmain   = lsci_t * shock_t * dist_min_norm_nonmain,
      indice2_nonmain    = lsci_t * shock_t * export_proxy_nonmain
    )
  panel_list_nonmain[[as.character(t)]] <- tmp
}
panel_data_nonmain <- bind_rows(panel_list_nonmain)
setDT(panel_data_nonmain)
panel_data_nonmain[, names(panel_data_nonmain)[grepl("\\.y$", names(
    panel_data_nonmain))] := NULL]
x_cols_nonmain <- names(panel_data_nonmain)[grepl("\\.x$", names(
    panel_data_nonmain))]
setnames(panel_data_nonmain, old = x_cols_nonmain, new = gsub("\\.x$"
    , "", x_cols_nonmain))

#index
years <- unique(LSCI_annual$year)
adm2s  <- unique(coastal_admin2_master_nonmain$ADM2_NAME)

panel_skeleton_nonmain <- expand.grid(ADM2_NAME = adm2s, year = years
    )

final_regression_data_nonmain <- merge(panel_skeleton_nonmain, LSCI_
    annual, by = "year", all.x = TRUE)
final_regression_data_nonmain <- merge(final_regression_data_nonmain,
     ContPortThroughput_long[, c("year","shock")], by = "year", all.x
    = TRUE)
final_regression_data_nonmain <- merge(final_regression_data_nonmain,
     nightlight_dt, by = c("ADM2_NAME","year"), all.x = TRUE)
final_regression_data_nonmain <- merge(final_regression_data_nonmain,
     conflicts_aggregated, by = c("ADM2_NAME","year"), all.x = TRUE)
final_regression_data_nonmain <- merge(
  final_regression_data_nonmain,
  st_drop_geometry(coastal_admin2_master_nonmain[, c("ADM2_NAME","
      dist_min_port_nonmain","dist_min_norm_nonmain","export_proxy_
      nonmain")]),
  by = "ADM2_NAME", all.x = TRUE
)
setDT(final_regression_data_nonmain)
final_regression_data_nonmain[is.na(n_conflicts), n_conflicts := 0]

# Coerenza con la definizione usata per i MAIN
```

```
epsilon <- 1e-6

#LOG nightlights for both index
#trade ports
final_regression_data_main <- final_regression_data_main %>%
  mutate(log_nightlight = log(nightlight_mean + 1))  # +1 per evitare log(0)
#dummy high
final_regression_data_main <- final_regression_data_main %>%
  group_by(year) %>%
  mutate(q3_log_nl = quantile(log_nightlight, 0.75, na.rm = TRUE)) %>%
  ungroup() %>%
  mutate(high_nl = ifelse(log_nightlight > q3_log_nl, 1, 0))
#interaction with index
final_regression_data_main <- final_regression_data_main %>%
  mutate(
    indice2_highNL = indice2_main * high_nl,
    indice2_lowNL  = indice2_main * (1 - high_nl)
  )
#non trade ports
final_regression_data_nonmain <- final_regression_data_nonmain %>%
  group_by(year) %>%
  mutate(q3_log_nl = quantile(log_nightlight, 0.75, na.rm = TRUE)) %>%
  ungroup() %>%
  mutate(
    high_nl = ifelse(log_nightlight > q3_log_nl, 1, 0),
    indice2_highNL_nonmain = indice2_nonmain * high_nl,
    indice2_lowNL_nonmain  = indice2_nonmain * (1 - high_nl)
  )
###regression####
#trade port
#baseline model
model_base <- feols(
  n_conflicts ~ indice2_main + log_nightlight | ADM2_NAME + year,
  data = final_regression_data_main
)
summary(model_base)
#interaction with high and low nightlights
model_interaction2 <- feols(
  n_conflicts ~ indice2_highNL + indice2_lowNL + log_nightlight | ADM2_NAME + year,
  data = final_regression_data_main
)
summary(model_interaction2)

##falsification test
#baseline model
final_regression_data_nonmain <- final_regression_data_nonmain %>%
  mutate(indice2_total_nonmain = indice2_highNL_nonmain + indice2_lowNL_nonmain)
model_sum_nonmain <- feols(
  n_conflicts ~ indice2_total_nonmain + log_nightlight | ADM2_NAME + year,
  data = final_regression_data_nonmain, cluster = ~ ADM2_NAME
)
```

```
700 model_sum_nonmain
701 model_interaction2_nonmain <- feols(
702   n_conflicts ~ indice2_highNL_nonmain + indice2_lowNL_nonmain + log_
        nightlight | ADM2_NAME + year,
703   data = final_regression_data_nonmain, cluster = ~ ADM2_NAME
704 )
705 model_interaction2_nonmain
706
707
708 ####Heterogeneity####
709 #geographical
710 adm <- coastal_admin2_master_main %>%
711   st_make_valid() %>%
712   st_transform(32631) %>%
713   mutate(centroid_m = st_point_on_surface(geometry))%>%
714   st_transform(4326) %>%
715   mutate(
716     lon = st_coordinates(centroid_m)[,1],
717     lat = st_coordinates(centroid_m)[,2]
718   )
719
720 # 2) Nord/Sud "coast-aware"
721 fit_coast <- lm(lat ~ poly(lon, 2, raw = TRUE), data = adm)
722 adm <- adm %>% mutate(
723   lat_pred = predict(fit_coast, newdata = data.frame(lon = lon)),
724   macro_NS = if_else(lat > lat_pred, "Nord", "Sud")
725 )
726
727 # 3) Est/Ovest
728 lon_med <- median(adm$lon, na.rm = TRUE)
729 adm <- adm %>% mutate(macro_EW = if_else(lon <= lon_med, "Ovest", "
       Est"))
730
731 #check rapido
732 print(table(adm$macro_NS, adm$macro_EW))
733
734 # 4) Join with the panel
735 adm_u <- adm %>% st_drop_geometry() %>%
736   dplyr::select(ADM2_CODE, macro_NS, macro_EW) %>%
737   mutate(ADM2_CODE = as.character(ADM2_CODE))
738
739 final_regression_data_main <- final_regression_data_main %>%
740   mutate(ADM2_CODE = as.character(ADM2_CODE)) %>%
741   dplyr::select(-matches("^macro_"), -matches("^indice2_(nord|sud|est
        |ovest)$"), everything()) %>%
742   left_join(adm_u, by = "ADM2_CODE")
743
744 #check
745 print(table(final_regression_data_main$macro_NS, final_regression_
       data_main$macro_EW))
746 #regression
747 mod_NS <- feols(
748   n_conflicts ~ i(macro_NS, indice2_main) + log_nightlight | ADM2_
        NAME + year,
749   data = final_regression_data_main, cluster = ~ ADM2_NAME
750 )
751 mod_EW <- feols(
```

```
752   n_conflicts ~ i(macro_EW, indice2_main) + log_nightlight | ADM2_
          NAME + year,
753   data = final_regression_data_main, cluster = ~ ADM2_NAME
754 )
755 summary(mod_NS); summary(mod_EW)
756
757 #sectorial
758 chem_first <- c("LAGOS", "TIN CAN ISLAND", "WARRI")
759 veg_first <- c("PORT HARCOURT", "ONNE", "CALABAR")
760
761 port_meta <- tibble::tibble(
762   MAIN_PORT_ = c(veg_first, chem_first),
763   port_sector = c(rep("Vegetables-first", length(veg_first)),
764                   rep("Chemicals-first", length(chem_first)))
765 )
766
767 ports_tagged <- ports_main_proj %>%
768   left_join(port_meta, by = "MAIN_PORT_")
769
770 adm2_cent <- coastal_admin2_master_main %>%
771   st_transform(st_crs(ports_tagged)) %>%
772   mutate(centroid = st_centroid(geometry)) %>%
773   dplyr::select(ADM2_NAME, geometry = centroid)
774
775 nearest_id <- st_nearest_feature(adm2_cent, ports_tagged)
776
777 adm2_port <- adm2_cent %>%
778   mutate(
779     nearest_port = ports_tagged$MAIN_PORT_[nearest_id],
780     port_sector  = ports_tagged$port_sector[nearest_id]
781   ) %>%
782   st_drop_geometry()
783
784 #join with the panel
785 df_sector <- final_regression_data_main %>%
786   left_join(adm2_port[, c("ADM2_NAME","nearest_port","port_sector")],
          by = "ADM2_NAME") %>%
787   mutate(
788     port_sector = ifelse(is.na(port_sector), "Unknown", port_sector)
789   )
790
791 m_sector <- feols(
792   n_conflicts ~ i(port_sector, indice2_main) + log_nightlight |
793     ADM2_NAME + year,
794   data = df_sector,
795   cluster = ~ ADM2_NAME
796 )
797 summary(m_sector)
798
799 ##trade flow
800 import_ports <- c("ONNE", "CALABAR")
801 export_ports <- c("LAGOS", "TIN CAN ISLAND", "PORT HARCOURT", "WARRI"
      )
802
803 port_flow <- tibble::tibble(
804   MAIN_PORT_ = c(import_ports, export_ports),
805   flow_group = c(rep("Import-oriented", length(import_ports)),
```

```
                  rep("Export-oriented", length(export_ports)))
)

ports_tagged2 <- ports_main_proj %>% left_join(port_flow, by = "MAIN_
    PORT_")

nearest_id2 <- st_nearest_feature(adm2_cent, ports_tagged2)
adm2_flow <- adm2_cent %>%
  mutate(flow_group = ports_tagged2$flow_group[nearest_id2]) %>%
  st_drop_geometry()

df_flow <- final_regression_data_main %>%
  left_join(adm2_flow[, c("ADM2_NAME","flow_group")], by = "ADM2_NAME
      ")

m_flow <- feols(
  n_conflicts ~ i(flow_group, indice2_main) + log_nightlight |
    ADM2_NAME + year,
  data = df_flow, cluster = ~ ADM2_NAME
)
summary(m_flow)

##tribes
tribes <- read_sf("tribalmap.shp"); tribes
st_crs(tribes)
st_crs(nig)
tribes_wgs84 <- sf::st_transform(tribes, sf::st_crs(admin2_in_coastal
    _admin1))
nt <- st_intersection(
  st_make_valid(tribes_wgs84),
  st_make_valid(admin2_in_coastal_admin1)
)

# weight the tribal units
nt <- nt %>%
  mutate(
    area_int = as.numeric(st_area(geometry))
  )


stopifnot(all(c("ADM2_NAME") %in% names(nt)))
tribe_var <- "TRIBE_CODE"

#tribal shares
tribal_shares <- nt %>%
  st_drop_geometry() %>%
  group_by(ADM2_NAME, !!sym(tribe_var)) %>%
  summarise(area = sum(area_int, na.rm = TRUE), .groups = "drop") %>%
  group_by(ADM2_NAME) %>%
  mutate(
    area_tot = sum(area, na.rm = TRUE),
    share    = ifelse(area_tot > 0, area / area_tot, NA_real_)
  ) %>%
  ungroup()

# indexes
tribal_index_admin2 <- tribal_shares %>%
```

```
860   group_by(ADM2_NAME) %>%
861   summarise(
862     # quota della trib  dominante
863     tribal_share_dom = max(share, na.rm = TRUE),
864     # fractionalization = 1 - sum(s_i^2)
865     tribal_frac      = 1 - sum(share^2, na.rm = TRUE),
866     .groups = "drop"
867   )
868
869 #regression
870 df <- final_regression_data_main %>%
871   left_join(tribal_index_admin2, by = "ADM2_NAME") %>%
872   mutate(
873
874     idxL_dom = indice2_lowNL  * tribal_share_dom,
875     idxH_dom = indice2_highNL * tribal_share_dom,
876     idxL_frac = indice2_lowNL  * tribal_frac,
877     idxH_frac = indice2_highNL * tribal_frac
878   )
879
880 #dominance
881 m_tr_dom <- feols(
882   n_conflicts ~ indice2_lowNL + indice2_highNL + idxL_dom + idxH_dom
        + log_nightlight |
883     ADM2_NAME + year,
884   data = df, cluster = ~ ADM2_NAME
885 )
886
887 #fractionalization
888 m_tr_frac <- feols(
889   n_conflicts ~ indice2_lowNL + indice2_highNL + idxL_frac + idxH_
        frac + log_nightlight |
890     ADM2_NAME + year,
891   data = df, cluster = ~ ADM2_NAME
892 )
893
894 summary(m_tr_dom)
895 summary(m_tr_frac)
896 #water
897 r <- rast(paste0(getwd(), "/wat/faowat00.tif"))
898
899 nig <- st_read("nga_polnda_adm2_1m_salb.shp")
900 nig <- st_transform(nig, crs(r))
901 r_crop <- crop(r, vect(nig))
902 r_mask <- mask(r_crop, vect(nig))
903 #mean
904 vals <- terra::extract(r_mask, vect(nig), fun=mean, na.rm=TRUE)
905
906 # dataframe
907 theme1_dt <- data.table(
908   ADM2_NAME = nig$ADM2_NAME,
909   water_avail = vals[,2]
910 )
911
912 #merge with panel
913 final_regression_data_main <- merge(
914   final_regression_data_main,
```

```
  theme1_dt,
  by="ADM2_NAME",
  all.x=TRUE
)

#regression
df_water <- final_regression_data_main %>%
  mutate(high_water = ifelse(water_avail.x > median(water_avail.x, na
      .rm=TRUE), 1, 0))


df_water <- final_regression_data_main %>%
  mutate(
    high_water = ifelse(
      water_avail > median(water_avail, na.rm = TRUE), 1, 0
    )
  )


q <- unique(quantile(final_regression_data_main$water_avail, probs =
    c(.333,.666), na.rm=TRUE))

df_water <- final_regression_data_main %>%
  mutate(
    water_terc = ntile(water_avail, 3),   # 1 = basso, 2 = medio, 3 =
        alto
    water_terc = factor(water_terc, labels = c("Bassa","Media","Alta"
        ))
  )


m_water_terc <- feols(
  n_conflicts ~ indice2_main * i(water_terc, ref = "Bassa") + log_
      nightlight |
    ADM2_NAME + year,
  data = df_water,
  cluster = ~ADM2_NAME
)

summary(m_water_terc)
```

Bibliography

ACLED. (2025). *Armed conflict location & event data project (acled) dataset*. https://acleddata.com/

Amodio, F., Baccini, L., Chiovelli, G., & Di Maio, M. (2024). Trade liberalization, economic activity and political violence in the global south: Evidence from ptas. *Economic Policy*, 39(118), 275–322.

Amodio, F., Baccini, L., & Di Maio, M. (2021). Security, trade, and political violence. *Journal of the European Economic Association*, 19(1), 1–37.

Anderson, J. E., & van Wincoop, E. (2003). Gravity with gravitas: A solution to the border puzzle. *American Economic Review*, 93(1), 170–192. https://doi.org/10.1257/000282803321455214

Autor, D. H., Dorn, D., & Hanson, G. H. (2013). The china syndrome: Local labor market effects of import competition in the united states. *American economic review*, 103(6), 2121–2168.

Ayesu, E. K., & Boateng, K. A. B. (2024). Estimating the impact of container port throughput on employment: An analysis for african countries with seaports. *Journal of Shipping and Trade*, 9(1), 8.

Berman, N., & Couttenier, M. (2015). External shocks, internal shots: The geography of civil conflicts. *Review of Economics and Statistics*, 97(4), 758–776.

Berman, N., Couttenier, M., Rohner, D., & Thoenig, M. (2017). This mine is mine! how minerals fuel conflicts in africa. *American Economic Review*, 107(6), 1564–1610.

Bertoni, E., Di Maio, M., Molini, V., & Nistico, R. (2019). Education is forbidden: The effect of the boko haram conflict on education in north-east nigeria. *Journal of Development Economics*, 141, 102249.

Brückner, M., & Ciccone, A. (2010). International commodity prices, growth and the outbreak of civil war in sub-saharan africa. *The Economic Journal*, 120(544), 519–534.

Burke, M., Hsiang, S. M., & Miguel, E. (2015). Climate and conflict. *Annu. Rev. Econ.*, 7(1), 577–617.

Campbell, B. (2010). Revisiting the reform process of african mining regimes. *Canadian Journal of Development Studies/Revue canadienne d'études du développement*, 30(1-2), 197–217.

Chen, X., & Nordhaus, W. D. (2011). Using luminosity data as a proxy for economic statistics. *Proceedings of the National Academy of Sciences*, 108(21), 8589–8594.

Collier, P., & Hoeffler, A. (1998). On economic causes of civil war. *Oxford economic papers*, 50(4), 563–573.

Collier, P., & Hoeffler, A. (2004). Greed and grievance in civil war. *Oxford economic papers*, 56(4), 563–595.

Dix-Carneiro, R., & Kovak, B. K. (2023). *Globalization and inequality in latin america* (tech. rep. No. w31459). National Bureau of Economic Research.

Dube, O., & Vargas, J. F. (2013). Commodity price shocks and civil conflict: Evidence from colombia. *Review of Economic Studies*, 80(4), 1384–1421.

Ducruet, C., Itoh, H., & Merk, O. (2014). *Time efficiency at world container ports* (Discussion Paper). International Transport Forum, OECD.

Esteban, J., Mayoral, L., & Ray, D. (2012). Ethnicity and conflict: An empirical study. *American Economic Review*, 102(4), 1310–1342.

FAO & IIASA. (2020). *Global agro-ecological zones (gaez v4) data portal*. Food, Agriculture Organization of the United Nations (FAO), and International Institute for Applied Systems Analysis (IIASA).

Fearon, J. D. (2003). Ethnic and cultural diversity by country. *Journal of economic growth*, 8(2), 195–222.

Fearon, J. D., & Laitin, D. D. (2003). Ethnicity, insurgency, and civil war. *American political science review*, 97(1), 75–90.

Fugazza, M., & Hoffmann, J. (2017). Liner shipping connectivity as determinant of trade. *Journal of Shipping and trade*, 2(1), 1.

Goldberg, P. K., & Pavcnik, N. (2007). Distributional effects of globalization in developing countries. *Journal of economic Literature*, 45(1), 39–82.

Goldsmith-Pinkham, P., Sorkin, I., & Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8), 2586–2624.

Henderson, J. V., Storeygard, A., & Weil, D. N. (2012). Measuring economic growth from outer space. *American economic review*, 102(2), 994–1028.

International Monetary Fund & University of Maryland. (2025, October 13). *Portwatch*. https://portwatch.imf.org/pages/port-monitor

Islam, Z. (2024). Measuring regional inequality using nightlight satellite data and population density for nigeria. *Human Geographies: Journal of Studies & Research in Human Geography*, 18(1).

Li, X., Zhou, Y., Zhao, M., & Zhao, X. (2020). Harmonization of dmsp and viirs nighttime light data from 1992–2021 at the global scale. *Scientific Data*, 7, 168.

Martin, P., Mayer, T., & Thoenig, M. (2008). Civil wars and international trade. *Journal of the European Economic Association*, 6(2-3), 541–550.

McGuirk, E., & Burke, M. (2020). The economic origins of conflict in africa. *Journal of Political Economy*, 128(10), 3940–3997.

Michalopoulos, S., & Papaioannou, E. (2013). Pre-colonial ethnic institutions and contemporary african development. *Econometrica*, 81(1), 113–152.

Michalopoulos, S., & Papaioannou, E. (2018). Spatial patterns of development: A meso approach. *Annual Review of Economics*, 10(1), 383–410.

Mishra, V. K., Dutta, B., Goh, M., Figueira, J. R., & Greco, S. (2021). A robust ranking of maritime connectivity: Revisiting unctad's liner shipping connectivity index (lsci). *Maritime Economics & Logistics*, 23(3), 424–443.

Montalvo, J. G., & Reynal-Querol, M. (2005). Ethnic polarization, potential conflict, and civil wars. *American economic review*, 95(3), 796–816.

Nigerian Ports Authority. (2025). *Ports and terminals*. https://nigerianports.gov.ng/ports/

Obi, C. I. (2010). Oil extraction, dispossession, resistance, and conflict in nigeria's oil-rich niger delta. *Canadian Journal of Development Studies/Revue canadienne d'études du développement*, 30(1-2), 219–236.

Raleigh, C., Linke, A., Hegre, H., & Karlsen, J. (2010). Introducing ACLED: An armed conflict location and event dataset. *Journal of Peace Research*, 47(5), 651–660.

Rauschendorfer, J., & Shepherd, B. (2022). Trade, conflict and informality: Evidence from the south sudanese civil war. *The World Economy*, 45(3), 867–884.

Storeygard, A. (2016). Farther on down the road: Transport costs, trade and urban growth in sub-saharan africa. *The Review of economic studies*, 83(3), 1263–1295.

UNCTAD. (2012). *The unctad and workport models of port development: Evolution or revolution*. United Nations Conference on Trade and Development. Geneva. https://unctad.org/system/files/official-document/dtltlb2011d1_en.pdf

UNCTAD. (2025a). *Liner shipping connectivity index (lsci)*. https://unctadstat.unctad.org

UNCTAD. (2025b). *Unctadstat - container port throughput*. https://unctadstat.unctad.org